SORTING CITIES OUT: HARMONIZING THE MEASURES OF SEGREGATION

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Abstract. Segregation scores in the social sciences come in two main families: unevenness scores like dissimilarity or entropy, which look at the makeup of the units, and spatial autocorrelation scores like Moran's I, which see how units are arranged. We reorganize the scores and propose an attractive summary through a weighted exposure metric we call Capy—on toy grids Capy behaves like I, while on real cities it behaves like unevenness. Our main result explains why: Capy splits exactly as a product of two factors, an *intensity* score that tracks unevenness and a *layout* score that tracks autocorrelation. In U.S. racial demographics over the last five censuses, tract-based intensity has fallen while layout sorting has held or risen, and Capy follows the product. Capy amounts to a neighbor correlation score, generalizing assortativity statistics from network science.

1. Multiplicity of scores, and a riddle

Is racial segregation rising or falling in America? The answer appears to depend on the metric, and the favored metrics from different fields disagree.

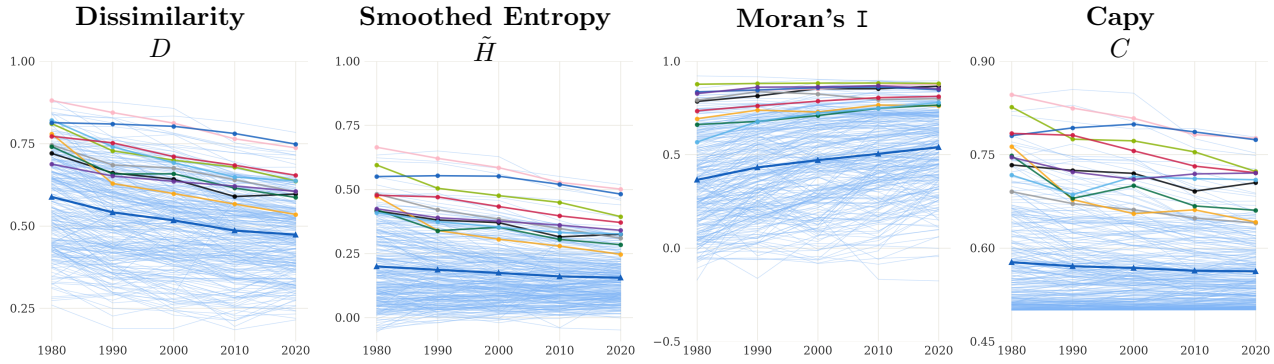


Figure 1. White-Black segregation on the level of metropolitan areas over time, highlighting the 10 most populous metro areas in colored lines, along with all metro areas with over 100,000 population and appearing in all decades ($N = 280$).

Sociologists have traditionally emphasized indices of unevenness (like D); entropy-based measures (like H and $\tilde{H}$) appear across the social sciences and statistical physics; geographers and spatial statisticians often focus on spatial autocorrelation (like I); and we will define C within a network science tradition that studies the tendency of similar types to be neighbors. Though they capture different features, all are meant to assess the degree of separation versus mixing. Applied to the same metropolitan populations, however, they sometimes announce opposite trends.

Figure 1 shows this divergence when several scores are computed for Black/White segregation in U.S. metropolitan areas for Census years from 1980 through 2020.[1] **Dissimilarity** declines

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Following the convention in mathematics, author order is alphabetical.

[1]We use Black/White segregation as our principal longitudinal case because it is the historically central U.S. comparison and admits a clean two-group interpretation; see also Appendices F and G for more generality.

substantially, reproducing a well-established finding in the segregation literature [Iceland et al., 2002, Logan and Stults, 2011, Firebaugh and Farrell, 2016]. **Entropy** measures give the same account, whether used in a classic "aspatial" form like Theil's H or smoothed to $\tilde{H}$ in the manner of Reardon and O'Sullivan [2004]. **Moran's** `I`—the canonical measure of spatial autocorrelation in social science—trends up, in sharp contrast.[2] This divergence appears both in the ten largest metros and across a balanced panel of metros observed throughout the study period. A similar split trend was flagged by Lloyd [2015], comparing D and `I` for England and Wales. Catney [2017] explicitly advocates 'considering unevenness and clustering concurrently,' documenting falling D alongside rising `I` for most minority ethnic groups from 2001 to 2011.

To probe the differences, we survey the segregation scores and formulate a field-bridging metric that we call **Capy**, importing assortativity ideas from network science through a construction that adapts them to gradated data (shares rather than types). Viewed on an auxiliary person-level graph, C encodes the Pearson correlation between group labels at the endpoints of a random directed edge. As we will show, C exactly combines the two kinds of score that Lloyd and Catney wanted side by side, but it does not see the global share ρ of a subgroup X; we therefore propose the pair (C, ρ) as a concrete and interpretable two-dimensional measurement framework.

By creating synthetic data and compiling a large historical dataset, we can isolate conditions for agreement. On simple grids with binary unit assignment, Capy tracks with Moran, assigning high values to grids with large single-color clusters, intermediate values to irregular mixtures, and distinguishing checkerboards as "anti-segregated." But with real-world data, the agreement dissolves. In that setting, nearly all of the menagerie of classical scores give the same account as Capy, while Moran goes its own way.

The mismatch poses a measurement riddle. The measures respond to distinct features of spatial organization, but these features tend to move together in elementary examples. Why have they moved apart in real cities? A real and independently measurable phenomenon in the geography of race gives a concrete explanation for the empirical disagreement: spatial diffusion of minority groups. Over the study period, racial and ethnic minority populations became less concentrated, transitioning from higher shares in fewer tracts to more widespread and decentralized patterns. This transformation reflects many processes, including enforcement of laws against housing discrimination, suburbanization of minority populations, and demolition of major public-housing developments built during the optimistic mid-century boom years. Diffusion changes the intensity of local racial separation even when the broad spatial pattern remains recognizable: a historically Black region, for example, may retain a Black character and a surrounding gradient while becoming less compositionally extreme. Evenness, entropy, and neighbor association all register this reduction in extremity or same-group exposure. Moran's `I`, because it standardizes spatial covariance by the total variation in tract composition, can largely divide out the change. It may remain high, or even increase, as the underlying differences between neighboring populations diminish.

The remainder of the paper develops this account. Section 2 defines Capy and places it alongside the other scores, introducing a reweighting scheme that accounts for both spatial relations and population distribution to allow readings at the level of individuals rather than geographical units. We then derive an exact factorization of Capy into two terms, which we call *intensity* (resembling D and H) and *layout* (resembling `I`): how much unit shares deviate from the group's overall level, and how those shares are arranged on the map. Section 3 uses these tools to identify mechanisms that push the scores apart. A controlled diffusion process reproduces the disagreement in Figure 1, and neighborhood-level examples let us recognize real-world diffusion in metropolitan census data. Finally, Section 4 revisits the Moran scatterplot [Anselin, 1996], a very successful interpretive device where `I` is the slope of a fit line, to extend the meaningful readings of its geometry.

[2]Other widely used spatial-association statistics include Geary's c, the Getis–Ord family, and join-count statistics for categorical data. All are used mainly as tests against a random-permutation null; Moran's `I` is the one most often also read as a top-line comparative index.

The individual-level lens organizes a crowded field of segregation metrics, and pays off three ways. Capy becomes a correlation of group membership between neighbors. That correlation is exactly intensity times layout. And all three quantities can be read off a single weighted scatterplot, in terms of the average members of each group.

2. Deeper comparison of scores

2.1. **Constructing a graph.** We treat an area to be studied as a graph on n nodes with demographic data attached (see Appendix E for details). Geographical units such as census tracts partition the region, and the *dual graph* has a node for each unit and an edge between units that are adjacent; A denotes its adjacency matrix. Suppose we study two population types X and Y, such as Black and White residents. At node i we record the integer counts x_i and y_i and the total population t_i; this means $t_i = x_i + y_i$ when the two types exhaust the population, which is the standing assumption for the main presentation.[3] Collect the data into vectors $\mathbf{x}$, $\mathbf{y}$, $\mathbf{t}$. Totals over the jurisdiction are $\bar{x} = \sum_i x_i$ and $T = \sum_i t_i$, and the global share of type X is $\rho = \bar{x}/T$. Since we typically focus on the group in the numerical minority, $0 < \rho \leq 1/2$ as we develop the mathematics. (When we move to real cities, some will be majority-minority.) The share of type X people at node i is $\rho_i = x_i/t_i$ which forms vector $\boldsymbol{\rho}$.

2.2. **The exploded graph and the Capy score.** We now introduce a second graph as a thinking tool for extending network definitions to shares—we will set up the mathematics so that we do not need to build this graph to perform computations. The goal is to measure the extent to which people of type X tend to live next to other people of type X rather than next to people of type Y, counting both the within-unit adjacencies and the adjacencies between neighboring units. To make this precise we associate to the dual graph a massively disaggregated auxiliary graph that we call the *exploded graph*: replace each node i with a complete graph on t_i vertices, exactly x_i of which are of type X, and if $i \sim j$ in the dual graph, include all $t_i t_j$ edges between the two cliques. This graph has one vertex per person and is helpful to explain what the following quantities count.

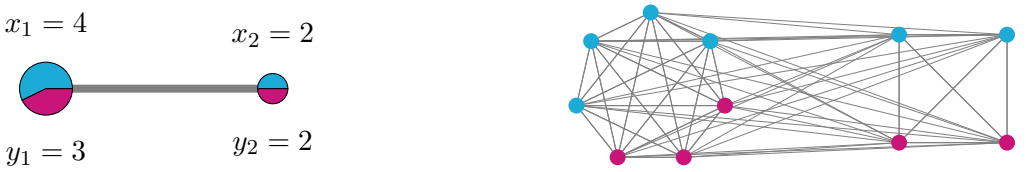


Figure 2. This figure shows the *exploded graph* associated to an initial graph with $\mathbf{x} = (4, 2)$, $\mathbf{y} = (3, 2)$, and no other type of population. We see that an edge in the unit-based dual graph becomes a join of two complete graphs in the auxiliary graph. In this case, the exploded graph has $\langle \mathbf{x}, \mathbf{y} \rangle = 30$ edges between different-colored nodes, 15 edges between X nodes, and 10 edges between Y nodes, making 55 edges in all. The proportion of X population overall is $\rho = 6/11$.

We will introduce a *spatial weight matrix* W (assumed symmetric with nonnegative entries) that is interpreted as giving us a weighting factor telling us how related a pair of units (i, j) are. For most purposes we will use $W = I + A$, which means that people are regarded as "neighbors" to others in their own unit (represented by the identity matrix I) and the geographically adjacent units (represented by the network's adjacency matrix A). The choice of spatial weight matrix W controls the notion of neighbors, or "local environment," in the graph.[4]

[3] We also assume $t_i > 0$ for all i, and enforce it in real data (Appendix E).

[4] For example, we can interpolate between within-unit and adjacent-unit weighting with $W_\lambda = I + \lambda A$. Standard scoring uses $\lambda = 1$, and $\lambda = 0$ gives a within-unit variant. Another choice that is popular in some geography settings is to take all units next to all others, but with decaying weights as the centroid distances grow.

Then we write $\langle \mathbf{u}, \mathbf{v} \rangle_W = \mathbf{u}^\top W \mathbf{v}$, or just $\langle \mathbf{u}, \mathbf{v} \rangle$ when W is understood. This means that $\langle \mathbf{x}, \mathbf{y} \rangle$ is the number of edges of the exploded graph joining an X person to a Y person, and $\langle \mathbf{x}, \mathbf{x} \rangle$ is twice the number joining two X people.[5] So among the half-edges starting at an X, the share that lead to another X is $\langle \mathbf{x}, \mathbf{x} \rangle / (\langle \mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle)$; we call this the *$X$-skew*, and define the Y-skew likewise. This is the probability that a randomly chosen neighbor pair $X\bullet$ is in fact XX. The Capy score is defined as the average of the two skews,

$$C := \frac{1}{2} \left[\frac{\langle \mathbf{x}, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle} + \frac{\langle \mathbf{y}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle} \right].$$

This is closely related to the family of *exposure* scores for segregation, but avoids centering one group's view by taking an average of the skew from the X and the Y point of view.

If every unit has the global composition, $\mathbf{x} = \rho\, \mathbf{t}$, then $\langle \mathbf{x}, \mathbf{x} \rangle = \rho^2 \langle \mathbf{t}, \mathbf{t} \rangle$ and $\langle \mathbf{x}, \mathbf{y} \rangle = \rho(1-\rho) \langle \mathbf{t}, \mathbf{t} \rangle$, which means $C = 1/2$ exactly. This holds for any graph, any W, and any population vector, and it gives C a consistent baseline. More generally, writing $a = \langle \mathbf{x}, \mathbf{x} \rangle$, $b = \langle \mathbf{x}, \mathbf{y} \rangle$, $c = \langle \mathbf{y}, \mathbf{y} \rangle$, the ordered pairs of neighbors in the exploded graph form a symmetric 2×2 table $\left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$ by type, sometimes called a contingency table. Its correlation coefficient (Pearson ϕ) is

$$\phi = \frac{ac - b^2}{(a+b)(b+c)}.$$

We have $\phi = 1$ when neighbors always match, $\phi = 0$ when knowing one neighbor's type tells you nothing about the other's, and $\phi < 0$ when neighbors differ more often than chance would predict. Since $\frac{a}{a+b} + \frac{c}{b+c} = 1 + \phi$, we get $C = (1+\phi)/2$, which means our average of skews is also an endpoint correlation in the graph, affinely scaled to lie in $[0, 1]$ instead of $[-1, 1]$. For a binary node attribute, the correlation ϕ is exactly Newman's assortativity coefficient r [Newman, 2003], so Capy has an exact interpretation as an assortativity statistic for the exploded graph.

2.3. **The menagerie of scores.** We start with three standard "evenness" indices that consider only compositions within units, but not layout across units. To define them, we start with two simple uncertainty measurements: $h(s) = -s \log_2 s - (1-s) \log_2 (1-s)$ (binary Shannon entropy) and $j(s) = 2s(1-s)$ (sometimes called a diversity index or Gini–Simpson index). Suppose a population has a share s of individuals of the X type. The uncertainty score h comes from information theory: the most efficient binary encoding of a long string of group types takes $h(s)$ bits per person. Even more simply, $j(s)$ is the chance that two people drawn independently from the population have different types. Over $0 \le s \le 1$, both are maximized at $s = 1/2$ and drop to zero uncertainty at $s = 0, 1$. Then we can write

$$D = \frac{\sum_i \frac{t_i}{T} |\rho_i - \rho|}{2\, \rho(1-\rho)}, \qquad H = \frac{h(\rho) - \sum_i \frac{t_i}{T} h(\rho_i)}{h(\rho)}, \qquad R = \frac{j(\rho) - \sum_i \frac{t_i}{T} j(\rho_i)}{j(\rho)}.$$

This D is the dissimilarity index of Duncan and Duncan, H is Theil's information (entropy) index, and R is the relative diversity index of Reardon and Firebaugh (which for two groups equals the "variance ratio" used since the 1950s) [Duncan and Duncan, 1955, Theil, 1972, Reardon and Firebaugh, 2002, James and Taeuber, 1985]. Each is zero when every unit matches the global composition and one when every unit is 100% solidly of a single type.

All three compare the composition of individual units to the global composition, averaged over people. D is the share of X people who would have to move, with Y people staying in place, to

[5] Twice, because each such edge contributes two half-edges based at type-X people. This points to a choice for the self-pairing convention: the diagonal term $W_{ii} x_i^2$ pairs each of the x_i people in unit i with themselves, whereas we might not want to count a person as their own neighbor. We keep the self-pairs in order to get cleaner mathematics (Remark 3 below); the discrepancy is negligible for tracts, but must be handled carefully for small examples like unit-population lattices, as in Appendix A.2.

make every unit match the global composition. H and R are the fraction of uncertainty about a person's type that is removed by knowing their unit, with uncertainty measured by h or j.

In a classic sociology paper, Reardon and O'Sullivan [2004] perform between-unit smoothing for all three the same way: replace the unit share ρ_i by the share of the unit's *local environment*,

$$\tilde{\rho}_i = \frac{(W\mathbf{x})_i}{(W\mathbf{t})_i},$$

the composition of the population within reach of unit i as defined by W. Reardon–O'Sullivan write $\tilde{D}$, $\tilde{H}$, $\tilde{R}$ for the smoothed forms. With $W = I$ the local environment is the unit itself and the within-unit indices are recovered; with $W = I + A$ it is the unit together with its neighbors in the dual graph. Their program aimed to free segregation scores from unit-dependence; we take up that aspiration, and the relationship of Capy to their framework, in Appendix A.2.

The most widely used statistic that sees the local environment via W is *Moran's* `I`, which is a dominant choice among geographers. Unlike the evenness indices, `I` depends on the *layout* of units—between-unit variation rather than within-unit variation is the fundamental quantity being measured. Write $\bar{\rho} = \frac{1}{n}\sum_i \rho_i$ for the average of the unit shares (which in general differs from the global share ρ, since each unit counts equally regardless of population), and $w = \sum_{i,j} W_{ij}$. Then

$$\mathtt{I}_W = \mathtt{I}(\boldsymbol{\rho}, W) = \frac{n}{w} \cdot \frac{\sum_{i,j} W_{ij}\,(\rho_i - \bar{\rho})(\rho_j - \bar{\rho})}{\sum_i (\rho_i - \bar{\rho})^2} = \frac{n}{w} \cdot \frac{\mathbf{v}^\top W \mathbf{v}}{\mathbf{v}^\top \mathbf{v}},$$

where $\mathbf{v}$ is the centered vector $\boldsymbol{\rho} - \bar{\rho}\mathbf{1}$. The usual choice is $W = A$ or the row-standardization that normalizes every row to sum to one. With $W = A$, Moran compares the average product of deviations across adjacent units to the average squared deviation at a unit, so it is near 1 when the spatial arrangement includes large clusters of units far above or far below average ρ_i, and near -1 for a checkerboard (because neighbors have opposite signs so contribute negatively to the total). The final expression recognizes `I` as a Rayleigh quotient, opening up a Fourier interpretation that suggests maximization at slowly-oscillating distributions (clusters) and minimization at fast-oscillating distributions (checkerboards). This spectral/harmonic connection is developed further in Duchin et al. [2023].

Looking back, the *numerator* of each evenness index (D, H, R) measures the spread of the unit shares about the center ρ (as an absolute deviation, an entropy gap, or a squared deviation, respectively).[6] By contrast, the *denominator* of `I` is a squared deviation about a slightly different center. Thus spread is what the evenness indices measure and what Moran's `I` divides out. The next section makes the alignment precise.

2.4. **A factorization theorem: intensity and layout.** In this section we show that with careful weighting that takes into account W and $\mathbf{t}$, we can obtain a clean statement that shows that Capy's correlation factors cleanly as an intensity (within-unit compositions) times layout (between-unit arrangement). (Figure 3 will help to visualize the meanings of the factors, and how they interact in the scores.) All of the reweighted quantities defined below are interpretable as statistics of a single random experiment: choose an ordered pair of neighboring individuals (q, q') in the exploded graph at random, with pairs weighted by W. The pairs with their individuals in units i, j have total weight $\omega_{ij} = W_{ij}\, t_i t_j$. Summing over j, the pairs starting in unit i have total weight

$$m_i = \sum_j \omega_{ij} = t_i\,(W\mathbf{t})_i,$$

which can be thought of as a reweighted degree of node i that sees both its population and the population of W-neighbors. (That unit has t_i residents and each of them has $(W\mathbf{t})_i$ generalized neighbors.) The total weight in the system is $M = \sum_i m_i = \langle \mathbf{t}, \mathbf{t} \rangle$.

[6] Since $\sum_i \frac{t_i}{T}\rho_i = \rho$, expanding j turns the numerator of R into $2\sum_i \frac{t_i}{T}(\rho_i - \rho)^2$, which is why R is also called the variance ratio.

So q lies in unit i with probability m_i/M, and q, q' are simultaneously in units i, j with probability ω_{ij}/M. This means that overall, across all units, q is of type X with probability $\rho_W := \sum_i (m_i/M)\,\rho_i = \langle \mathbf{x}, \mathbf{t}\rangle / \langle \mathbf{t}, \mathbf{t}\rangle$, so that ρ_W represents the X share of the reweighted population. When every unit has the same number of generalized neighbors $(W\mathbf{t})_i$, then ρ_W is just ρ; in general the two differ only when X people tend to live in more or less populous local environments than Y people. Write $\delta_i = \rho_i - \rho_W$ to recenter the population share vector at this baseline ρ_W.

Layout. The X share of the unit containing q is a random quantity with mean ρ_W and variance $\mathrm{Var}_W = \sum_i (m_i/M)\,\delta_i^2$. Since ω is symmetric, the share of the unit containing q' has the same distribution, and the covariance of the two is $\mathrm{Cov}_W = \sum_{i,j} (\omega_{ij}/M)\,\delta_i \delta_j$. We define the *layout score* $L_W = \mathrm{Cov}_W / \mathrm{Var}_W$, the correlation between the compositions of two neighbors' units. It is positive when neighbors tend to live in units of similar composition and negative when they tend to live in units of opposite composition. It is undefined only when $\mathrm{Var}_W = 0$, that is, when every unit has the same share.

Proposition 1 (Bounds on layout score)**.** Let W be symmetric with nonnegative entries. Then $-1 \le L_W \le 1$ for every population vector $\mathbf{t}$ and share vector $\boldsymbol{\rho}$. Furthermore, there are clean graph-specific lower bounds and null values in terms of the rowsums $w_i = \sum_j W_{ij}$ and total weight $w = \sum_i w_i$.

(1) If $W_{ii} \le {}^{w_i}/_2$ for some i, then $L_W \ge 2\left(\min_i \frac{W_{ii}}{w_i}\right) - 1$;
(2) If people of each type are placed uniformly at random on the units, then $\mathbb{E}[L_W] \approx \frac{\mathrm{tr}\, W}{w}$.

For $W = I + A$, this simplifies further in terms of max and average degree: the lower bound is $\frac{1-d_{\max}}{1+d_{\max}}$ and the neutral baseline is $\frac{1}{1+d_{\mathrm{avg}}}$.

Comparison with Moran's I. Moving the constant into the denominator, Moran's `I` can be written as

$$\mathtt{I}_W = \frac{\sum_{i,j} W_{ij}\,(\rho_i - \bar\rho)(\rho_j - \bar\rho)}{\sum_i \frac{w}{n}\,(\rho_i - \bar\rho)^2}, \qquad \text{while} \qquad L_W = \frac{\sum_{i,j} \omega_{ij}\,(\rho_i - \rho_W)(\rho_j - \rho_W)}{\sum_i m_i\,(\rho_i - \rho_W)^2}.$$

So L_W is Moran's `I` computed over person pairs instead of unit pairs, which calls for replacing each term with its weighted counterpart. If all units have equal population t and W has constant row sums k, then $\omega_{ij} = t^2 W_{ij}$, $m_i = kt^2 = t^2\, w/n$, and $\rho_W = \bar\rho = \rho$, so $L_W = \mathtt{I}_W$ exactly.

Reweighting to L_W also bounds the score across all graphs and distributions (Proposition 1). By contrast, $\mathtt{I}_W$ can fall outside $[-1, 1]$ on graphs with uneven degrees for the usual choices of spatial weight matrix (including A and $I + A$), and can be made arbitrarily large [Duchin et al., 2023].

Intensity. The axis of segregation measured by layout concerns arrangement or the gross pattern of units. Next we develop a corresponding tool in the evenness family. Var_W measures the unevenness of the shares ρ_i across units. For a given ρ_W it is largest when every unit is extreme with share 0 or 1, taking its maximum value $\rho_W(1 - \rho_W)$. We define the *intensity score* as the unevenness normalized by that maximum, $R_W = \dfrac{\mathrm{Var}_W}{\rho_W(1-\rho_W)}$. Moving terms around, we see that this is the counterpart to the compositional score R from above:

$$R = \frac{j(\rho) - \sum_i \frac{t_i}{T}\, j(\rho_i)}{j(\rho)}; \qquad R_W = \frac{j(\rho_W) - \sum_i \frac{m_i}{M}\, j(\rho_i)}{j(\rho_W)}.$$

When every unit has the same number of weighted neighbors, i.e., $(W\mathbf{t})_i$ is constant over i, we have $R_W = R$. Like R, the weighted intensity score does not depend on how units are arranged.

	Intensity R_W					Layout L_W					ϕ	
Metro	1980	1990	2000	2010	2020	1980	1990	2000	2010	2020	1980	2020
New York	.656	.673	.676	.645	.611	.855	.870	.885	.888	.896	.561	.547
Los Angeles	.702	.588	.577	.539	.472	.929	.934	.942	.944	.945	.652	.446
Chicago	.789	.737	.690	.631	.610	.878	.880	.893	.894	.905	.693	.552
Dallas	.642	.435	.377	.375	.338	.819	.817	.822	.860	.837	.526	.283
Houston	.609	.453	.479	.408	.383	.815	.794	.836	.821	.839	.496	.321
Washington	.552	.492	.466	.483	.485	.891	.903	.902	.906	.908	.491	.441
Philadelphia	.661	.644	.585	.533	.502	.859	.873	.875	.869	.880	.568	.442
Miami	.667	.530	.547	.515	.505	.651	.701	.784	.820	.831	.434	.419
Atlanta	.563	.517	.489	.437	.454	.829	.869	.900	.876	.901	.467	.410
Boston	.455	.404	.386	.352	.333	.837	.847	.836	.841	.841	.381	.280
Mean	.261	.221	.206	.186	.180	.459	.500	.528	.561	.594	.154	.125

TABLE 1. Intensity and layout factors of Black/White segregation, with $W = I + A$ on tract graphs of MSAs (mean over $N = 280$ metros). The product $\phi = R_W \cdot L_W$ is shown at the beginning and end of the time window. (Note that the mean ϕ is the mean of the product $R_W L_W$ across metros and not a product of the means.)

Theorem 2 (Factorization Theorem)**.** For every symmetric W with nonnegative entries and every population vector $\boldsymbol{\rho}$, the neighbor correlation factors exactly as a within-unit score (intensity) times a between-unit score (layout):

$$\phi = \frac{\mathrm{Cov}_W}{\rho_W(1-\rho_W)} = R_W \cdot L_W.$$

Proof. Since $R_W L_W = \mathrm{Cov}_W/\rho_W(1-\rho_W)$ by definition, it suffices to show ϕ equals the same quantity. With a, b, c as above, write each in terms of shares: $a = \sum_{i,j} \omega_{ij}\rho_i\rho_j$; $b = \sum_{i,j} \omega_{ij}\rho_i(1-\rho_j)$; $c = \sum_{i,j} \omega_{ij}(1-\rho_i)(1-\rho_j)$. Since $\sum_j \omega_{ij} = m_i$, we have $a+b = \langle \mathbf{x}, \mathbf{t}\rangle = M\rho_W$ and $b+c = M(1-\rho_W)$. Expanding the covariance gives $\mathrm{Cov}_W = \frac{a}{M} - \rho_W^2$. Substituting $c = M - 2M\rho_W + a$ yields $ac - b^2 = M(a - M\rho_W^2) = M^2\mathrm{Cov}_W$. Dividing by $(a+b)(b+c) = M^2\rho_W(1-\rho_W)$ gives the claim. □

In words, ϕ asks how well the type of one neighbor predicts the type of the other. The answer factors: how well a unit predicts the type of its residents (R_W), times how well the composition of one unit predicts the composition of a neighboring unit (L_W). This is visualized in Figure 3, where $-3/5 \leq L_W \leq 1$ by Proposition 1, with a non-sorted baseline close to $L_W = 1/5$.

Remark 3 (Self-pairing and quadratic forms)**.** This section makes heavy use of the fact that $\langle \mathbf{x}, \mathbf{x}\rangle_W = \mathbf{x}^\top W\mathbf{x}$ is a quadratic form, and so are Var_W and Cov_W in the share deviations δ. These have a counting interpretation that includes self-pairing. Adopting definitions based on quadratic forms has three consequences. First, scores can be computed with linear algebra rather than combinatorics, which is what makes them readily scalable. Second, this leads to the appearance of R rather than D or H in the factorization: of those, only R has a numerator that is a quadratic form in $\boldsymbol{\rho}$, so only R_W can cancel the denominator of L_W. Finally, the self-pairing is ultimately what allows person-level covariance and correlation to be calculated at the unit level: conditioned on the unit membership of q and q', their types τ and τ' are *independent*, with $\mathbb{P}(\tau = 1) = \rho_i$ and $\mathbb{P}(\tau' = 1) = \rho_j$, so by the law of total covariance $\mathrm{Cov}(\tau, \tau')$ equals $0 + \mathrm{Cov}(\rho_i, \rho_j) = \mathrm{Cov}_W$ (with the zero coming from independence). Without self-pairs, a within-unit pair is a draw without replacement, and the factorization picks up a correction of order $\sum_i W_{ii}t_i/M$.

Corollary 4. If all units have equal population and W has constant rowsum, then factorization is already exact with the classical variance ratio and Moran's $\mathtt{I}$; that is, $\phi = R \cdot \mathtt{I}_W$.

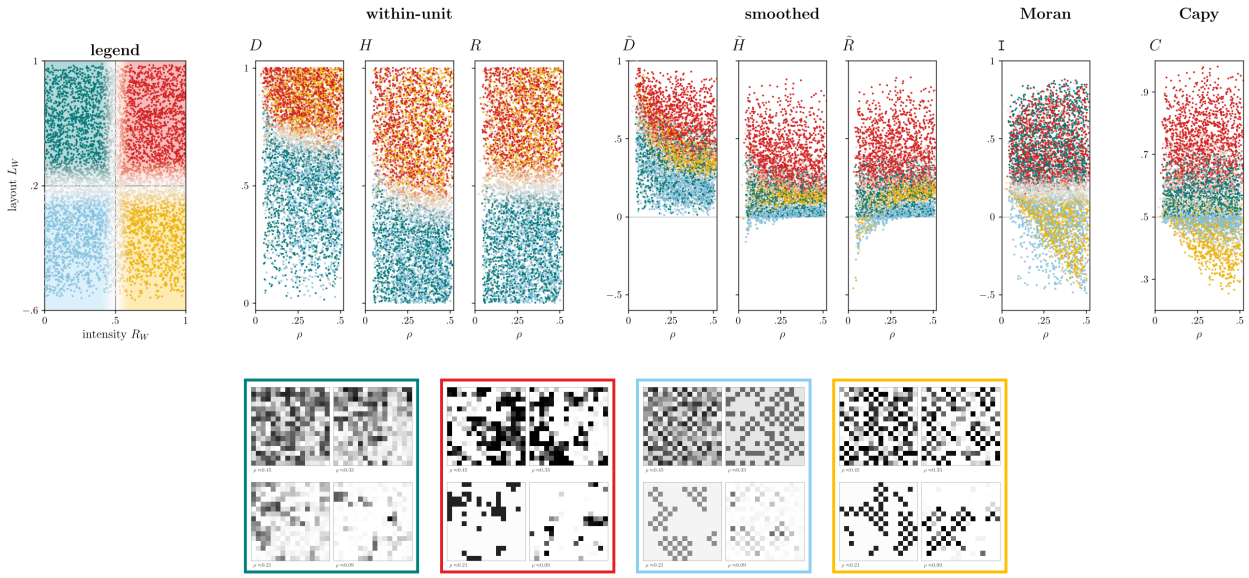


FIGURE 3. Illustrating factorization. Over 5000 synthetic 16×16 grids with unequal unit populations are colored by their quadrant in the (R_W, L_W) plane (with $W = I + A$ and coloring cuts at $R_W = .5$ and $L_W = .2$). The datapoints are plotted against ρ for each score. The within-unit scores register intensity alone, while Moran's `I` sees layout alone. The smoothed scores are partly successful at seeing both, but Capy does best at that task. The bottom row shows examples of grids near the center of each quadrant. (Also compare Figure 8.)

There is a clean way to see rigorously why R_W and L_W should be regarded as suitable coordinates. If we encode the distributional data in $\boldsymbol{\delta}$, the vector of differences of ρ_i from ρ_W, then (R, L) come from size and direction in a polar coordinate system for $\boldsymbol{\delta}$. To be precise, R_W is just a scaling of $\sum_i m_i \delta_i^2$, so it depends only on a norm for $\boldsymbol{\delta}$, and L_W is a ratio of two quadratic forms in $\boldsymbol{\delta}$, which cancels out the norm and depends only on the direction; they are truly independent coordinates.

3. Mechanisms for divergence

To return to the riddle: why, in real urban geography, is one family of scores dropping over the last five census timestamps, while another family is steady or rising? The Factorization Theorem explains the split: the evenness family measures intensity while autocorrelation sees only layout. We can then hypothesize that American segregation is easing on the intensity axis while remaining stubbornly high in terms of gross layout—and Table 1 confirms it. The product is giving us an interpretable indication of how those trends combine: a modest drop in neighbor correlation in New York, for example, and a much more pronounced drop in Dallas.

Some examples in urban history are suggestive of a precise mechanism for the split. In Chicago, for example, the Bronzeville neighborhood is an important and historic district on the city's South Side. Up to mid-century, Bronzeville contained the majority of Black residents of the city in just 2% of its land area.[7] After the mid-1940s, Bronzeville experienced a pattern of demographic shift that is aptly described as **one-sided diffusion**: the Black residential footprint dissolved into surrounding neighborhoods and Bronzeville's density dropped, but to this day no other population has entered to replace the Black families that moved out. On the other hand, many neighborhoods

[7]To be precise, the community areas called Douglas, Grand Boulevard, and Washington Park comprise 4.91 square miles out of about 227 in the city. By 1940 census numbers, the neighborhood held 73% of Black Chicagoans.

have experienced **two-sided diffusion**, in which one group spreads outward from an enclave while other groups move in. These are illustrated in the next section.

3.1. **Illustrative experiments.** To resolve the riddle, we look for a real-world process that pushes intensity and layout readings to diverge. Diffusion is a natural candidate. Figure 4 shows a series of experiments on grid-graphs which demonstrate a major difference in how these two families of scores respond to one-sided and two-sided diffusion.[8] In both cases, the intensity scores and Capy fall steadily. Moran's I has an initial increase in both, then stabilizes in one-sided diffusion and eventually drops in a noisy fashion under two-sided diffusion.

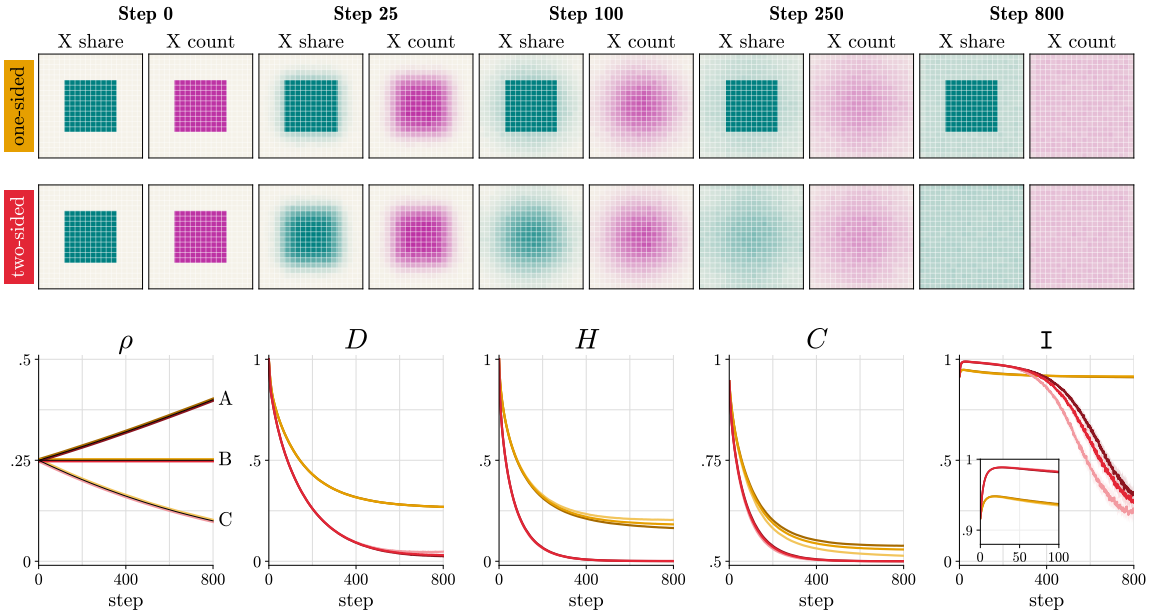


FIGURE 4. Six patterns of stochastic diffusion are shown here. Series 1 (amber traces) is one-sided diffusion, where group X individuals spread by a random walk process while Y individuals do not move; series 2 (red traces) applies random movement to both groups. For each series, there are A, B, and C variants in which the group X population level is rising, steady, or falling, respectively. The grid snapshots show X share (ρ_i) and X count ($\rho_i t_i$) in the steady variants (1B, 2B), while the trace plots show all six. 25 runs of 800 steps are conducted for each variant, and the median line is plotted, together with the 10th-90th percentile band, which is narrow enough to be hidden behind the median in most panels.

Diffusion softens sharp boundaries, damping fast-oscillating components first and slowly oscillating ones last. It therefore lowers intensity while raising layout sorting through killing local checkerboarding, like the corner of the inner enclave, which was solidly X with two neighboring units solidly Y. This means that an initially rising layout score is consistent with population-level mixing rather than evidence against it.

[8]The Figure 4 experiments are conducted in a 20×20 grid with 1000 people per cell; X is initialized in a central 10×10 block with only Y in the rest of the grid, so $\rho = 0.25$ at Step 0. To achieve rising or falling ρ, we implement birth-in-place or a survival probability for X population, respectively, tuned to double expected population over 800 steps in condition A or drop it to one-third in condition C. For the random walk itself, each movable individual independently shifts with probability 0.2 to one of its rook neighbors chosen uniformly, and otherwise stays in place.

3.2. **Observational data.** Finally, we turn to examples with White/Black segregation measurement in Chicago and Philadelphia to locate dispersion within the time range of our dataset (Figure 5).

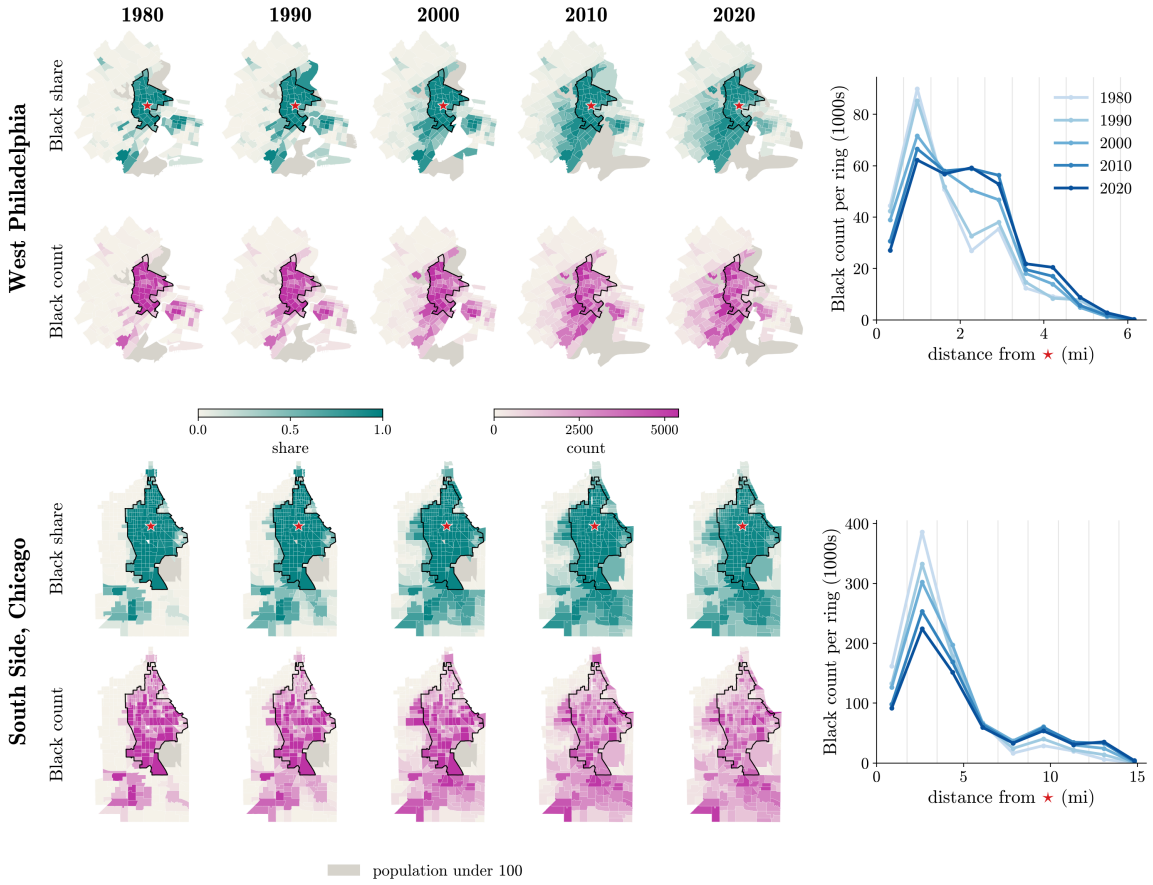


FIGURE 5. Diffusive patterns for Black population in West Philadelphia and the South Side of Chicago. In both examples, the high-share core becomes less concentrated while the Black residential footprint spreads outward over time.

Urban sociology distinguishes many kinds of turnover, not all of them diffusive. One classic account is *residential succession*: a subgroup's footprint spreads, and the boundary passes through a mixed stage before it turns over [Park et al., 1925, Taeuber and Taeuber, 1965].[9] Lower Manhattan's Chinatown grew this way into Little Italy after 1965 [Zhou, 1992], and Boston's Blue Hill Avenue corridor was Jewish before it was Black [Levine and Harmon, 1992]. *Blockbusting* is an accelerated version, in which real estate agents stoke panic-selling to speed a turnover, as in Baltimore's Edmondson Village from 1955 to 1965 [Orser, 1994]. *Gentrification* has a different modality: a more affluent (and generally Whiter) population moves in, and priced-out households are often scattered far away, as with many Latino residents of San Francisco's Mission District [Mirabal, 2009]. Finally, a disruptive event like the demolition of a public housing complex or a Chinatown district, or a coordinated displacement like the forced removal of Japanese Americans from Los Angeles in 1942, provokes a scatter that may or may not be accompanied by a corresponding influx. Each of these would have a different signature in intensity and layout scores.

[9]The Chicago School borrowed the term "invasion–succession" from plant ecology, but that may now read as jarring when describing neighborhood change.

4. Another view of factorization

Anselin [1996] made an influential observation about the geometry of spatial autocorrelation: with a row-standardized W, Moran's I is the slope of the least-squares line through the points $(\rho_i, (W\boldsymbol{\rho})_i)$, now called the *Moran scatterplot*. This lets us see what units drive the score and which are outliers, and the quadrants of the plot locate tracts that are like or unlike their neighbors. Capy has an analogous picture, and the whole factorization theorem appears in its geometry. Plot each tract at $(\rho_i, \tilde{\rho}_i)$, where $\tilde{\rho}_i = (W\mathbf{x})_i/(W\mathbf{t})_i$ is the group share of its local environment (replacing the average of the shares of neighboring tracts from the Anselin plot). Now fit a line to these datapoints by least squares with weights m_i. The slope of this line is exactly L_W, and the normal equations for the fit say that the line passes through two reference points: the weighted averages of $(\rho_i, \tilde{\rho}_i)$ over members of each group. The horizontal displacement between these points is R_W, and the vertical displacement is $\phi = 2C - 1$.

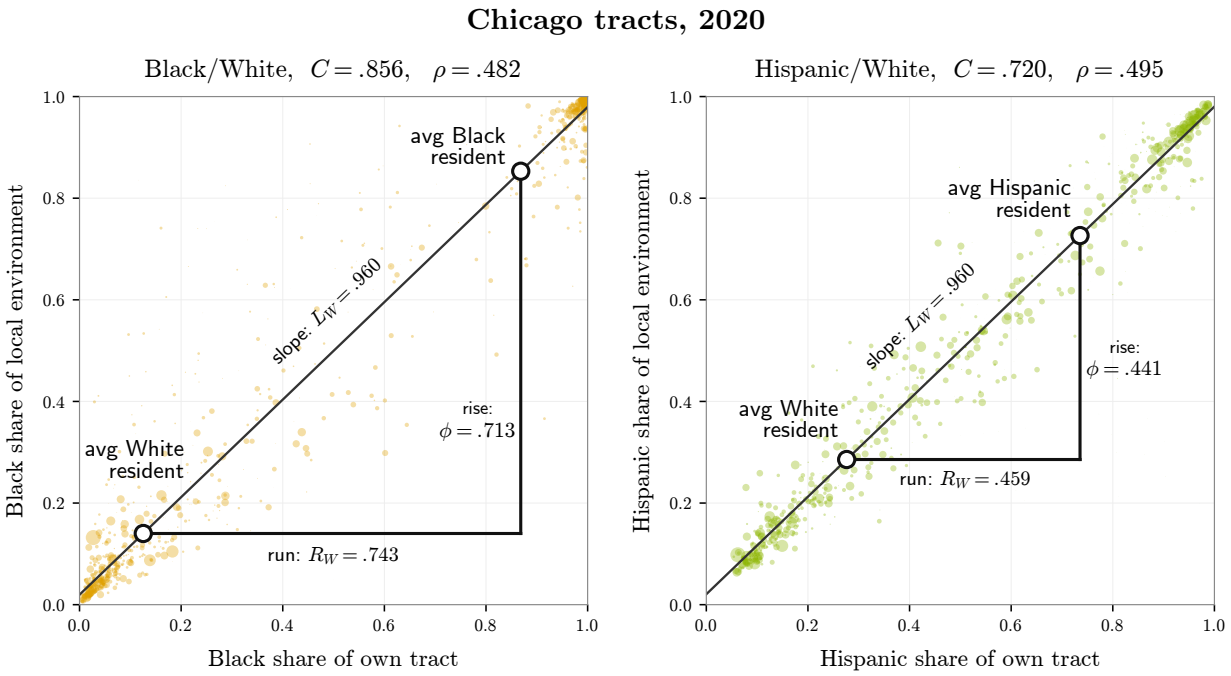


FIGURE 6. The scatterplot shows Black/White and Hispanic/White data in Chicago proper from 2020, with tracts sized by the weight $m_i = t_i(W\mathbf{t})_i$ for $W = I + A$. Points are placed at $(\rho_i, \tilde{\rho}_i)$ with $\tilde{\rho}_i = (W\mathbf{x})_i/(W\mathbf{t})_i$. The weighted least-squares fit line is shown; it exactly contains reference points showing the expected tract and environment shares for a weighted average over individuals from each group.

Since slope=rise/run, we can read the factorization theorem $\phi = R_W \cdot L_W$ as just rise=run×slope. Figure 6 shows this for Chicago in 2020. The two panels have essentially the same slope ($L_W = .960$), so the difference in segregation measured by Capy is entirely attributable to the greater variance measured by R_W. Spatial autocorrelation is the same, but very few tracts have a mix of Black and White residents, while the Hispanic/White tract compositions are still bimodal, but fill in more values in the intermediate range.

5. Conclusion

We have introduced Capy and presented it as a product of a layout score and an intensity score. We argue that two dimensions (C, ρ) are very expressive for understanding demographic patterning at the population scale. Why two dimensions and not one? Because ρ adds crucial context that can't be read off from C. Why two dimensions and not three? Since $C = (RL + 1)/2$, we could propose (R, L, ρ) as a three-dimensional summary instead. Indeed, this is a superior framework in terms of specificity. Nevertheless, $C = (1 + \phi)/2$ is an interpretable summary of the combined effect of R and L (see Table 1). This means that the two-dimensional summary (C, ρ) is preferable where parsimony matters. And when trends compete, the factorization can help disentangle their contribution.

More broadly, the development in this article suggests an updated set of axioms for measuring segregation. We state them for a measurement framework—a small collection of scores rather than a single scalar—because composition, polarization, and spatial sorting are distinct features whose effects can be separated. (Appendix B makes each axiom precise and compares them with axioms proposed by earlier authors.) Here, *measurement* means any quantity derived from the scores, and *reference values* may be numerical values or signs.

New axioms for measures of segregation.

Comp *Global composition is isolated.* Some measurement reports the overall subgroup share ρ, and no other score is determined by it: each takes a range of values at every ρ, with reference values that do not depend on ρ.

Pol *Polarization of shares is isolated.* Some measurement tracks how far unit compositions vary from the overall share, independent of rearrangements.

Sort *Sorting of units is isolated.* Some measurement tracks whether similar units sit next to each other, independent of the spread of unit shares.

Assoc *Association is visible.* Some measurement reports association between the group membership of an individual and the membership of their neighbors, with reference values for complete, absent, and reversed association.

Check *Checkerboarding is visible.* Some measurement distinguishes interleaved patterns from clustered ones, separated by a random-arrangement reference value.

Inv *Invariance.* No measurement depends on which group is labeled first (up to exchanging ρ for $1 - \rho$), on the overall population scale, on adding disjoint copies of a region, or on splitting a unit into two identical neighbors with the same composition and adjacencies.[10]

All six are upheld by (R_W, L_W, ρ), and the simpler (C, ρ) gets partial credit. (See Appendix B for a scorecard.)

The riddle that opened the paper is therefore not that the scores disagree, or even that they are answering different questions, but how. Reweighting to R_W and L_W harmonizes the metrics: they measure distinct features while their product gains a direct interpretation. Over the last five decennial censuses, the contrasts of racial segregation have softened in intensity, even as the gross spatial sorting has endured or ticked up. The first effect outweighs the second: race-based segregation persists, but race is now substantially less predictive of who lives next door.

[10]The splitting proviso brings to mind MAUP, or the *modifiable areal unit problem*, which geographers use to flag worries about unit-dependence of spatial statistics. We would prefer a degree of MAUP stability: limited change under more general merging and splitting of units. MAUP invariance is unattainable as a literal matter, so this stability should be judged qualitatively, as in Figure 15.

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Appendix A. A more comprehensive review of the segregation literature

A.1. **Segregation scores across fields.** Segregation scores as such have been developed largely independently in sociology and demography, geography and spatial statistics, and network science. Roughly, the first tradition measures how far unit compositions spread from the whole, the second measures how units are arranged, and the third measures who is tied to whom. More broadly, the phenomena behind the segregation measures in this paper go by an astonishing array of names across other fields, in addition to assortativity, exposure, isolation, entropy, and spatial autocorrelation discussed here. Network scientists have homophily and assortative mixing; physicists have ferromagnetic ordering. Economists have sorting, agglomeration or concentration, and fractionalization. Ecologists have (β-)diversity, patchiness, and the term segregation itself. Population geneticists use differentiation (F_{ST}), compositional variability, and isolation by distance.

We attempt an extremely succinct overview of the most related ideas here, noting which fields favor which scores, and then make connections to Capy.

Unevenness and entropy. The oldest and most widely used segregation scores compare the composition of each unit to the composition of the whole. Dissimilarity was consolidated as the standard by Duncan and Duncan [1955], ushering in what Massey and Denton [1988] call the "Pax Duncana," and it remains the workhorse of sociology and demography, including the Census Bureau's trend reports [Iceland et al., 2002]. The entropy index H was proposed by Theil for school segregation and is often preferred in economics and education research, largely because it extends to many groups and decomposes additively across levels of geography and organization [Theil, 1972, Reardon and Firebaugh, 2002]. The variance ratio R is the statistician's version (and matches the F_{ST} statistic from population genetics). R is the share of variance in a person's group membership explained by their unit, and it equals the isolation index normalized by global composition, $({}_X\tilde{P}^*_X - \rho)/(1-\rho)$ [James and Taeuber, 1985]. This normalization goes all the way back to Bell [1954], who read the isolation index as the probability that a group member's next meeting within their tract is with their own group, and normalized it so that groups of different sizes could be compared.

In the taxonomy of Massey and Denton, which organized twenty indices into five dimensions, D and H measure evenness while the isolation index (${}_X\tilde{P}^*_X$) and variance ratio (R) measure exposure. All of these scores average over people, center at ρ, and see only the unit each person lives in. They respond to how far unit shares spread from ρ and are blind to where the units sit, which is the checkerboard problem [White, 1983]. Massey and Denton's clustering dimension is the one that sees layout, and the index they recommend for it is White's spatial proximity score, discussed below. The surveys of James–Taeuber, Massey–Denton, and Reardon–Firebaugh, with their references, give a fuller picture of this literature.

Spatial autocorrelation. Geographers and spatial statisticians approach the same data with tools that focus on unit arrangement. They ask whether neighboring units resemble each other; this is the "contiguity problem," the geographer's name for the checkerboard problem [Massey and Denton, 1988]. The standard tool is Moran's `I`. Introduced as a test for spatial correlation among neighboring plots in agricultural field trials, `I` compares the covariance of unit shares across W-neighbors to the variance of unit shares [Moran, 1950]. Each unit counts once, and the shares are centered at their unweighted mean $\bar{\rho}$. Geary's c uses squared differences between neighbors instead. The Getis–Ord statistics measure local concentration of high values, and join counts handle units that are entirely one group. (Join counts are parallel to nearest-neighbor pair counts in plant ecology or bond counts for spins in statistical physics.) As noted in the introduction, these scores most often serve as tests against a random-permutation null, for example to detect clustering or to check model residuals.

Local environment. A third approach keeps the focus on people but replaces each person's unit with a neighborhood around them. White [1983] named the "checkerboard problem" and proposed a spatial proximity score called SP that compares the average relatedness of same-group pairs, under a distance-decay function, to that of all pairs. It is the index Massey and Denton recommend for clustering. Reardon and O'Sullivan [2004] rebuilt D, H, R, and exposure around a kernel-weighted local environment for each person. Reardon et al. [2008, 2009] then varied the kernel's radius and found that trends can differ by scale: from 1990 to 2000, Black–white segregation fell at small radii but was essentially unchanged at large ones. Chodrow [2017] writes H and several related indices as a Bregman information, a population-weighted average divergence of local compositions from the global one, so that they all inherit a chain-rule decomposition into between-region and within-region terms. He uses this to learn regions from the data instead of fixing them in advance, and he defines a local measure of spatial scale, so the scale of segregation can vary within a city rather than being set once for the whole region.

Networks. Network scientists often bring similar measurement goals to the study of graphs whose nodes are people or places and whose edges are ties or relationships. Newman's assortativity coefficient r compares the share of edges joining same-type nodes to what random mixing with the same type shares would give, so that it is 0 for random mixing and 1 when all edges are within a type [Newman, 2003]. It is routine in physics and network science and has recently been used directly as a segregation measure on population-scale social networks [Kazmina et al., 2024].

In economics, Echenique and Fryer [2007] start from the same quantity as Capy: for them, it is each person's share of contacts who belong to their own group. Where an isolation index averages these shares uniformly and Capy weights them by number of contacts, Echenique and Fryer weight them by the Perron eigenvector of the same-group contact matrix, so their spectral segregation index is its leading eigenvalue. This construction up-weights people deep inside same-group clusters. They apply it to school friendship networks and to residential data, treating census blocks within a kilometer of each other as neighbors and assigning each block its majority race.

How Capy fits. Each score above was designed with several implementation choices: how it averages (over individuals, over units, or over ordered/unordered pairs of either); what is taken as a center or baseline; whether it admits a choice of local environment W; and how it is normalized. From the usual network perspective using spatial graphs, the nodes are geographic units and the half-edges or edges are ordered or unordered pairs, respectively; one of the central suggestions of the present paper is that the exploded graph construction allows for individuals as nodes and pairs as edges. The other is aligning the choices so that averaging, centering, and normalization all make parallel use of the local environment W.

Capy was first proposed in a 2018 preprint [Alvarez et al., 2018] and is developed here as a bridging device between many viewpoints and literatures. It connects to most of the previous approaches. Its summands are built from join counts and can be read as probabilities that one step of a W-random walk stays within each group.

Notably, other authors have recognized that segregation is well described by pairing an intensity score with a layout score [Lloyd, 2015, Catney, 2017, Le Roux et al., 2017]. For example, Gemma Catney writes that "Taken together, D and `I` provide a composite measure of overall variation or spread (D) and the spatial configuration of values (`I`)." Our contribution is a careful choice of representatives from each family and a careful choice of weighting that turns them into a clean pair of axes (R_W, L_W) with an interpretable product ϕ.

Recovering earlier scores in special cases. Capy is a reweighted version of White's spatial proximity index SP and it is an interaction model in the sense of Bell. Our X-skew and Y-skew are reweighted versions of the isolation indices ${}_X\tilde{P}^*_X$ and ${}_Y\tilde{P}^*_Y$; and our layout score L_W is a reweighted Moran's `I`.

To be precise:

- [if $\rho_W = \rho$]: the spatial proximity index satisfies $SP = 2C$.
- [if $(Wt)_i$ is constant]: $\rho_W = \rho$ and $R_W = R$; isolation indices coincide with our skews so that ${}_X\tilde{P}^*_X = \rho + (1-\rho)\phi$ and ${}_Y\tilde{P}^*_Y = (1-\rho) + \rho\phi$. Each of those is individually ρ-dependent, but summing them cancels that out: $C = \frac{1}{2}\big({}_X\tilde{P}^*_X + {}_Y\tilde{P}^*_Y\big)$. Since exposure ${}_Y\tilde{P}^*_X$ satisfies ${}_Y\tilde{P}^*_X = 1 - {}_Y\tilde{P}^*_Y = \rho(1-\phi)$, this means $C = 1 - {}_Y\tilde{P}^*_X/(2\rho)$ under these conditions.
- [if all t_i are equal and W has constant rowsums $w_i = k$]: $L_W = I_W$, so $\phi = R \cdot I_W$.

The three conditions above are successively more specific (each implies the previous one). Two other notable cases:

- [if all $\rho_i \in \{0, 1\}$ and all t_i are equal]: $R_W = 1$, so $\phi = L_W$ and $C = (1 + r)/2$, where r is Newman's assortativity of the dual graph (holds exactly for $W = A$; for $W = I + A$, needs half-weight self-loops).
- [if $W = I$ and all t_i are equal]: $\phi = R_W$, so $C = (1 + R)/2$.

A.2. **A deeper dive into the work of Reardon–O'Sullivan.** Introducing the smoothed scores $\tilde{D}, \tilde{H}, \tilde{R}$ was part of a broader program to reduce the dependence of segregation scores on the choice of areal units. Reardon–O'Sullivan state their framework for continuous-density "surfaces" and a proximity kernel, and argue that this frees the measures from the modifiable areal unit problem. In practice, the inputs are unit counts (such as tract counts), the kernel is evaluated on a discrete support, and the local environment of a point is a weighted sum over nearby units—so in applications, the continuous-surface construction still amounts to a weighting matrix acting on discrete areal data. The dependence on discretization is transformed rather than eliminated.[11] We take the discrete setting as the starting point rather than as an approximation: the dual-graph neighborhood is the local environment, and the endpoint correlation ϕ is the fundamental measurement.

Reardon and O'Sullivan argue persuasively that much of what is meant by segregation can be captured in two quantitative dimensions, not one, and they introduce a schematic figure of four binary-colored arrangements on a triangular lattice, labeled so that the vertical axis runs from evenness to clustering and the horizontal from isolation to exposure. We use self plus nearest neighbors ($W = I + A$), which they endorse in a footnote, and extend each pattern periodically so that edge effects do not enter. They recommend $\tilde{H}$ and the White perspective on $\tilde{P}^*$, so we read the implied coordinates as $({}_Y\tilde{P}^*_X, \tilde{H})$, where X is the Black group and Y is the White group, noting that low $\tilde{H}$ is plotted higher.

Under that reading, the horizontal axis behaves as described, but the vertical axis does not. The SW panel is clustered by any measure ($\tilde{H} = 0.36$), but the SE panel, offered as the clustered-and-exposed paradigm, is the most even of the four ($\tilde{H} = 0.016$): its small triangular clusters give every Black dot a share of Black neighbors matching the global Black share, so the pattern is statistically indistinguishable from random allocation at the nearest-neighbor scale.

The upper panels raise a sharper point about checkerboarding. Reardon and O'Sullivan call a population "hyper-integrated" if individuals, on average, "experience greater diversity in their local environments than the diversity of the population as a whole," and state that a segregation index should then be negative. In a footnote they compute that in the NE pattern each White household's environment is 3/7 Black and each Black household's is 6/7 White, and expect a negative score. But their indices do not produce the intended negative value: $\tilde{H} = 0.070$, $\tilde{R} = 0.082$, and $\tilde{D}$, an absolute value, cannot be negative at all.

[11] Reardon's associated software package `seg` illustrates this point: the default is to use a lattice of points, receiving their data by proportional allocation from census units.

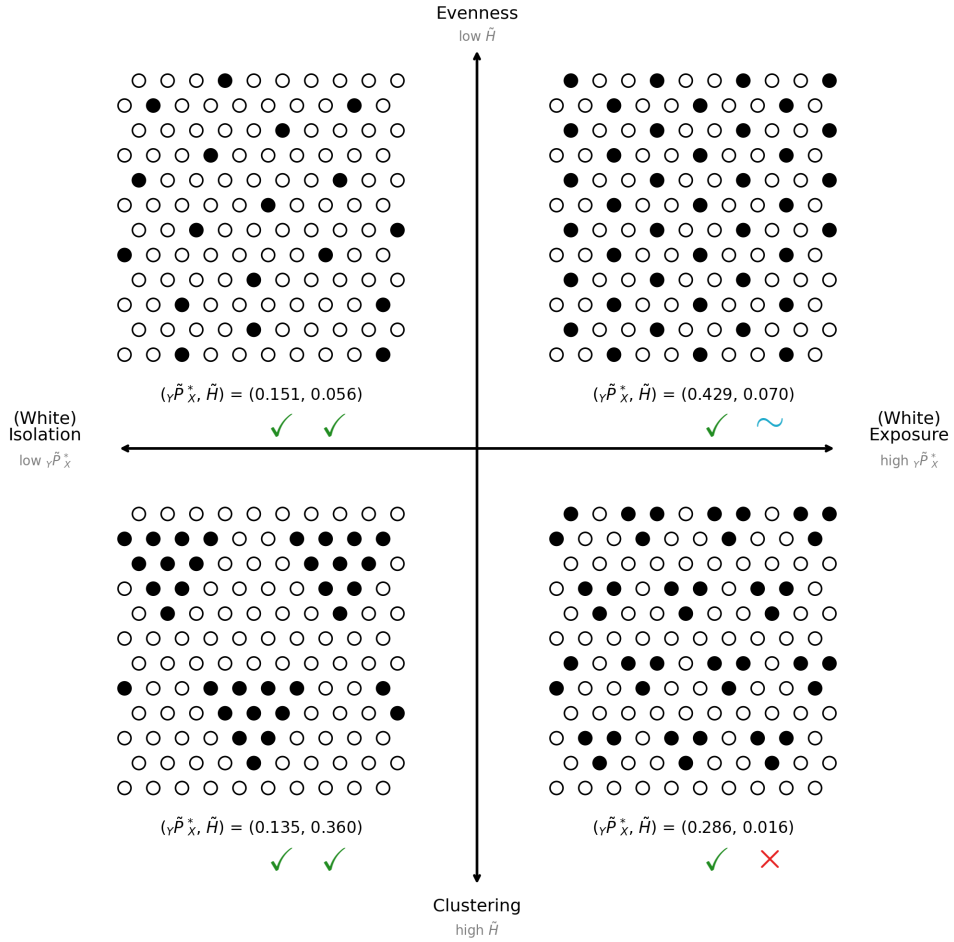


FIGURE 7. Reardon–O'Sullivan example, expanded, with marks indicating whether the scores order each pattern in the direction suggested by the schematic axes. $\tilde{H}$ measures the spread of smoothed local shares, so calling it "clustering" invites a category error: an intensity score risks being interpreted as a layout score. A smoothed intensity score cannot tell environments tilted toward one's own group from environments tilted away (NE panel), and it can't see clusters smaller than the kernel (SE panel). A layout score is needed for this purpose.

This can't be fixed by finding a new example. The smoothed indices can take negative values, but never for the reason they have in mind. By concavity, $\tilde{H}$ and $\tilde{R}$ can be negative only when the local shares $\tilde{\rho}_i$ fail to average to ρ, which happens when local environments vary in size—on an equal-population lattice, this happens *only* as a boundary effect. So on an equal-population lattice, no arrangement can receive a negative $\tilde{H}$ or $\tilde{R}$ for the substantive reason they describe. What the NE example does show is that each group has more neighbors from the other group than the overall composition would predict. That is *checkerboarding* (detected here by $\phi = -2/7$). Figure 8 shows the consequence on synthetic grids: checkerboards score near zero under the smoothed indices, alongside even and random patterns; the smattering of negative values comes from population variation and boundary effects.

A.3. **Two dimensions of segregation.** Finally, consider their two proposed dimensions of segregation: evenness (for which their preferred measure is $\tilde{H}$) and exposure/isolation (for which they prefer ${}_Y\tilde{P}^*_X$). To see a clean explanation of interdependence, consider the cousin $\tilde{R}$ in place of $\tilde{H}$

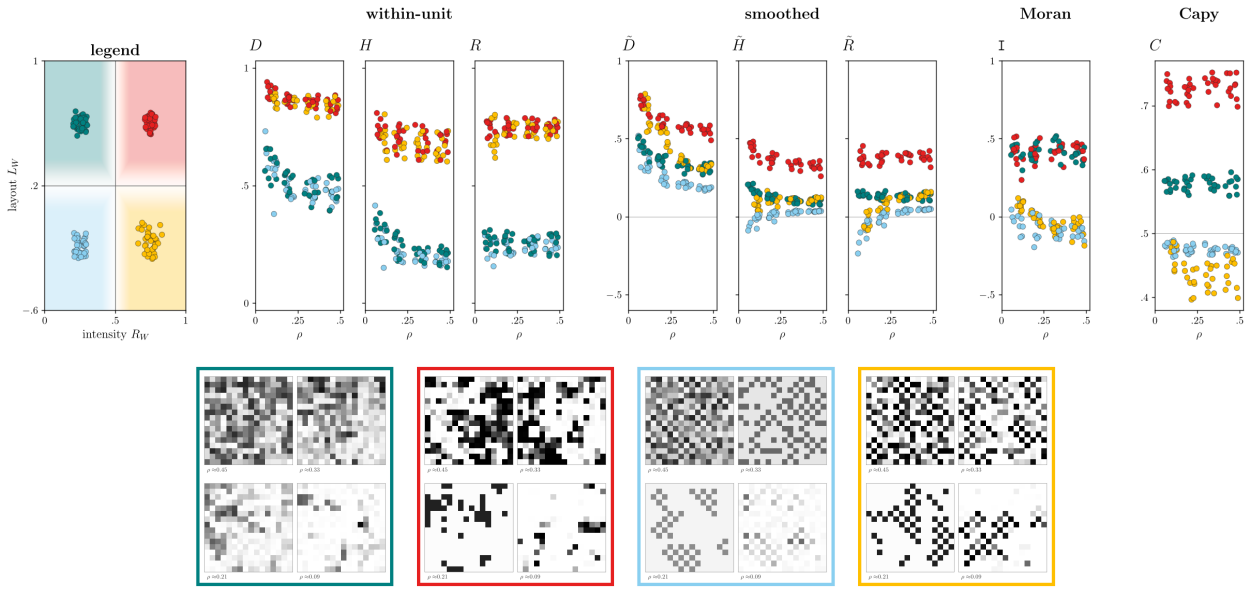


FIGURE 8. The grids of Figure 3, restricted to those near the center of each quadrant. This clearer look shows that the Reardon–O'Sullivan smoothed scores fail to distinguish checkerboards, which live in the high-intensity, negative-layout quadrant (yellow).

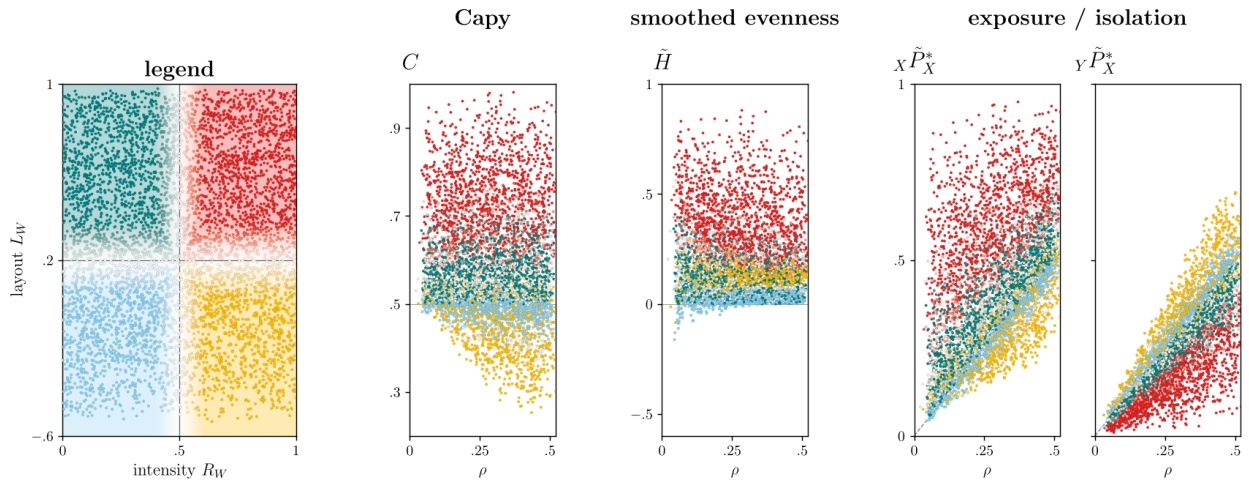


FIGURE 9. Reardon–O'Sullivan's preferred coordinates on the synthetic grids of Figure 3. The smoothed entropy index $\tilde{H}$ struggles to go negative and does not distinguish the checkerboard quadrant (yellow). The isolation and exposure indices are C sheared by ρ, so their no-pattern baseline moves with the composition.

and a normalized exposure $\frac{\rho - {}_Y\tilde{P}^*_X}{\rho}$. When every unit has the same local population (for instance, equal populations on a d-regular graph with $W = I + A$), local environments are $\tilde{\boldsymbol{\rho}} = P\boldsymbol{\rho}$ with $P = W/(d+1)$, and these reduce to

$$\frac{\rho - {}_Y\tilde{P}^*_X}{\rho} = RL, \qquad \tilde{R} = RL',$$

where $L = \mathtt{I}(\boldsymbol{\rho}; P)$ is a Moran-type score (coinciding with Layout in this setting) and $L' = \mathtt{I}(\boldsymbol{\rho}; P^2) = \frac{\|P\boldsymbol{\delta}\|^2}{\|\boldsymbol{\delta}\|^2}$ is a two-step Moran-type score. (The formula relating ${}_Y\tilde{P}^*_X$ and RL shows a sheared relationship, visible in Figure 9.) Cauchy–Schwarz gives $L^2 \le L'$; since $0 \le R \le 1$, we get

$$(RL)^2 \le RL'.$$

So one Reardon–O'Sullivan coordinate bounds the values attainable by the other, whereas the alternative coordinates (R, L) are independent (as discussed above at the end of Section 2.4). The same constraint holds approximately for general graphs and weights, and for $\tilde{H}$ in place of $\tilde{R}$, since $\tilde{H} \approx \tilde{R}/(2h(\rho)\ln 2)$ when the local shares $\tilde{\rho}_i$ are close to ρ.

Appendix B. Axiom systems

Starting with Duncan and Duncan [1955], criteria and axioms for segregation indices have proliferated, with at least James and Taeuber [1985], Reardon and Firebaugh [2002], Hutchens [2004], Frankel and Volij [2011] giving criteria for a single score. The idea that segregation is fundamentally multidimensional goes back at least to Massey and Denton [1988], and Reardon and O'Sullivan [2004] propose two segregation dimensions; in that spirit, we present axioms for a measurement *framework*, meaning a small collection of scores.

Deformations of the data. To study a tuple of scores in this vein requires identifying appropriate moves or transformations of a distribution on a graph. Concretely, we need

- a spatially weighted network of units (recorded by W);
- a polarization deformation P_s that fixes ρ, while spreading or concentrating unit shares monotonically in s;
- a rearrangement move that changes which unit compositions are paired, giving a *random-arrangement reference value* as the average over all rearrangements;
- a random placement of individuals on the units, holding the populations t_i fixed, giving a *random-placement reference value* as the expectation.

To test the scores in this paper, we use the polarization deformation $P_s(\rho_i) = \rho + s(\rho_i - \rho)$. This is linear interpolation: P_s slides from ρ to ρ_i (and beyond, until shares hit the end of their range). For rearrangement, we define a clean move on the pair weights $\omega_{ij} = W_{ij}t_it_j$. An elementary rearrangement is a weighted edge switch: for distinct i, j, k, ℓ and $0 \le \varepsilon \le \min(\omega_{ij}, \omega_{k\ell})$, set

$$\begin{aligned} \omega'_{ij} = \omega'_{ji} &= \omega_{ij} - \varepsilon, & \omega'_{k\ell} = \omega'_{\ell k} &= \omega_{k\ell} - \varepsilon, \\ \omega'_{ik} = \omega'_{ki} &= \omega_{ik} + \varepsilon, & \omega'_{j\ell} = \omega'_{\ell j} &= \omega_{j\ell} + \varepsilon, \end{aligned}$$

leaving all other entries unchanged. Such switches preserve every rowsum $m_i = \sum_j \omega_{ij}$, and hence preserve the endpoint distribution as well as ρ_W and R_W, while changing which unit compositions are paired. In other words, this implementation of rearrangement changes the coupling while preserving its marginals; it can be thought of as rewiring the edges. A random-arrangement reference value can be obtained by degree-preserving randomization generated by these switches.

Axioms (stated in terms of moves and deformations).

Comp The global shares $\{\rho, 1-\rho\}$ are read off directly. For every other coordinate, the reference values for uniform distribution and for random placement of individuals do not depend on ρ (for fixed t and W), and each coordinate takes a range of values at every $0 < \rho < 1$.

Pol Some coordinate changes monotonically along P_s and is fixed by rearrangements.

Sort Some coordinate is sensitive to rearrangement but invariant under P_s.

Assoc The coordinates determine an association of group labels across the coupling ω, with reference values for complete, absent, and reversed association.

CHECK Some coordinate falls on opposite sides of its random-arrangement reference value for interleaved and for clustered patterns.

INV Scores are unchanged by swapping the labels X and Y (up to $\rho \leftrightarrow 1-\rho$), by scaling all populations, by taking disjoint copies of a region, and by splitting a unit into two adjacent units with the same composition and same adjacencies.

Comparison with classical criteria. Other authors have discussed size invariance and symmetry in groups; these are covered by INV. Organizational equivalence is given a spatial interpretation in INV. Transfer and exchange axioms are conceptually incorporated into POL, with P_s as the coordinated transfer move. Hyper-integration is now subsumed in CHECK. ASSOC has no counterpart among the classical criteria, though the variance ratio R (isolation minus exposure) is the "aspatial" version of association.

On the other hand, the axiom sometimes called "composition invariance" is intentionally excluded here. It refers to scaling one group's counts by a factor of λ in each unit while leaving the other group unchanged; this is not readily interpretable in a residential segregation setting, and it is unclear what the desirable behavior might be for a metric under this kind of deformation. Similarly, sending some t_i to λt_i while holding its ρ_i fixed is a kind of move that would leave `I` invariant because it only sees unit shares, but is probably better regarded as a bug than a feature. (Scaling up a very unusual node should not leave segregation unchanged.)

Checklist for the axioms.

	COMP	POL	SORT	ASSOC	CHECK	INV
(D,ρ), (H,ρ)	$\sim$	✓	–	–	–	✓
(R,ρ)	✓	✓	–	–	–	✓
$(\mathtt{I},\rho)$	✓	–	✓	–	✓	$\sim$
$(D,\mathtt{I})$	–	✓	✓	–	✓	$\sim$
$({}_Y\tilde{P}^*_X,\tilde{H})$	–	–	–	–	$\sim$	–
(C,ρ)	✓	–	–	✓	✓	✓
(R_W,L_W,ρ)	✓	✓	✓	✓	✓	✓

TABLE 2. Which measurement frameworks satisfy which axioms.

The dissimilarity and entropy indices have uniform-distribution reference value 0, but their expected values under random placement of individuals depend on ρ, while those of R, `I`, R_W, L_W, and C do not vary with ρ, to leading order. (A related ρ–dependence of D and H is visible in Figure 8, where R is flat.) The pair $(D,\mathtt{I})$ has been proposed as a two-dimensional summary [Lloyd, 2015, Catney, 2017]; it separates polarization from sorting, but it does not report ρ, and supplying ρ would earn only partial credit, since the random-placement reference value of D depends on ρ.

Moran's `I` weights units equally, so splitting a unit changes it, while the other spatially weighted scores are preserved under splitting with $W = I + A$. Association is visible in (C,ρ) through $\phi = 2C - 1$ and in (R_W, L_W, ρ) through $\phi = R_W L_W$; Moran's `I` correlates unit shares rather than individuals. Reardon and O'Sullivan's $({}_Y\tilde{P}^*_X,\tilde{H})$ are both sensitive to the deformations and to rearrangement, but fail to register checkerboarding; exposure reads above its baseline for interleaved patterns, but that baseline is ρ itself, and exposure is not label-symmetric.

Multigroup extensions and stability under general redrawing of units are beyond our current scope, though one kind of merging is visualized in Figure 15.

Appendix C. Expanded proofs

Proposition 1 (Bounds on layout score)**.** Let W be symmetric with nonnegative entries. Then $-1 \le L_W \le 1$ for every population vector $\mathbf{t}$ and share vector $\boldsymbol{\rho}$. Furthermore, there are clean graph-specific lower bounds and null values in terms of the rowsums $w_i = \sum_j W_{ij}$ and total weight $w = \sum_i w_i$.

(1) If $W_{ii} \le w_i/2$ for some i, then $L_W \ge 2\left(\min_i \frac{W_{ii}}{w_i}\right) - 1$;

(2) If people of each type are placed uniformly at random on the units, then $\mathbb{E}[L_W] \approx \frac{\operatorname{tr} W}{w}$.

Proof. Recall that $\delta_i = \rho_i - \rho_W$ and $m_i = t_i(W\mathbf{t})_i$. The normalizing constant M cancels in the ratio, so

$$L_W = \frac{\sum_{i,j} W_{ij} t_i t_j \delta_i \delta_j}{\sum_i m_i \delta_i^2}.$$

The bounds $-1 \le L \le 1$. Since $\sum_j W_{ij} t_j = (W\mathbf{t})_i$, expanding the squares gives

$$\frac{1}{2}\sum_{i,j} W_{ij} t_i t_j (\delta_i \pm \delta_j)^2 = \sum_i m_i \delta_i^2 \pm \sum_{i,j} W_{ij} t_i t_j \delta_i \delta_j.$$

The left side is nonnegative because W has nonnegative entries. Taking the minus sign gives $L_W \le 1$, and taking the plus sign gives $L_W \ge -1$.

The graph-specific lower bound. Let $c = \min_i W_{ii}/w_i$. The hypothesis that $W_{ii} \le w_i/2$ for some i says exactly that $c \le 1/2$. It is convenient to change variables to $u_i = t_i \delta_i$, which turns the numerator and denominator of L_W into

$$N = \sum_{i,j} W_{ij} u_i u_j, \qquad D = \sum_i \frac{(W\mathbf{t})_i}{t_i}\, u_i^2 = \sum_{i,j} W_{ij} \frac{t_j}{t_i}\, u_i^2.$$

We want to show $N \ge (2c-1)D$, or equivalently $N + (1-2c)D \ge 0$.

Separate the diagonal terms $i = j$ from the off-diagonal terms. On the diagonal, N and D each contribute $\sum_i W_{ii} u_i^2$. Off the diagonal, group the ordered pairs (i,j) and (j,i) together, using the symmetry of W. This gives

$$N + (1-2c)D = (2-2c)\sum_i W_{ii} u_i^2 + \sum_{i<j} W_{ij}\, E_{ij},$$

where for each pair of distinct units we have set

$$E_{ij} = 2u_i u_j + (1-2c)\Big(\frac{t_j}{t_i} u_i^2 + \frac{t_i}{t_j} u_j^2\Big).$$

The E_{ij} are the only parts of the expression that use $\mathbf{t}$ terms, and they can be removed with the AM–GM inequality:

$$\frac{t_j}{t_i} u_i^2 + \frac{t_i}{t_j} u_j^2 \;\ge\; 2|u_i|\,|u_j|.$$

Since $1 - 2c \ge 0$, this gives

$$E_{ij} \;\ge\; 2u_i u_j + 2(1-2c)|u_i u_j| \;\ge\; -2|u_i u_j| + 2(1-2c)|u_i u_j| \;=\; -4c\,|u_i u_j| \;\ge\; -2c\,(u_i^2 + u_j^2),$$

where the last step uses $2|u_i u_j| \le u_i^2 + u_j^2$.

Now sum over pairs. Each unit i appears in the pairs $\{i,j\}$ for $j \ne i$, so

$$\sum_{i<j} W_{ij}\,(u_i^2 + u_j^2) = \sum_i u_i^2 \sum_{j \ne i} W_{ij} = \sum_i (w_i - W_{ii})\, u_i^2.$$

Putting this together,

$$N + (1-2c)D \;\geq\; \sum_i u_i^2 \Big[(2-2c)W_{ii} - 2c\,(w_i - W_{ii})\Big] \;=\; 2\sum_i u_i^2\,\big(W_{ii} - c\,w_i\big) \;\geq\; 0,$$

where the final inequality is the definition of c. Dividing by $D > 0$ gives $L_W \geq 2c - 1$. □

The hypothesis $W_{ii} \leq w_i/2$ is needed: if $W = \left[\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right]$ with $a > b$, then $2c - 1 = \frac{a-b}{a+b} > 0$, but $L_W \to 0$ as $t_1/t_2 \to 0$.

Derivation of null value. Place N people uniformly at random on the units, each independently of type X with probability ρ. Conditional on the resulting populations $\mathbf{t}$, the X-count in each unit is binomial, so $\rho_W \approx \rho$ and to leading order the shares ρ_i are uncorrelated with $\mathrm{Var}(\rho_i) \approx \frac{\rho(1-\rho)}{t_i}$. Only the diagonal terms survive in expectation, so

$$\mathbb{E}\Big[\sum_{i,j} W_{ij} t_i t_j \delta_i \delta_j\Big] \approx \rho(1-\rho)\sum_i W_{ii} t_i, \qquad \mathbb{E}\Big[\sum_i m_i \delta_i^2\Big] \approx \rho(1-\rho)\sum_i (W\mathbf{t})_i = \rho(1-\rho)\sum_i w_i t_i,$$

using the symmetry of W in the last step. The ratio of these is $\sum_i W_{ii} t_i / \sum_i w_i t_i$. Both numerator and denominator are linear in $\mathbf{t}$, and $\mathbb{E}[t_i] = N/n$ for every unit, so averaging over placements gives $\operatorname{tr} W/w$. □

Remark 5 (The sorting null with and without population distribution)**.** If instead the unit populations $\mathbf{t}$ are held fixed and only the X/Y labels are placed at random, the same computation shows that the null value $\mathbb{E}[L_W]$ has the leading term

$$\frac{\sum_i t_i W_{ii}}{\sum_i t_i w_i},$$

which is the probability that a random person, choosing a W-related neighbor, chooses one in their own unit. (That makes sense as a baseline for a layout score, which asks if your neighbor's unit composition is like yours.) This is arguably the natural baseline for a particular city, since it uses that city's population distribution but varies its X/Y composition within units. It agrees with $\operatorname{tr} W/w$ when populations are equal, or more generally when W_{ii}/w_i is constant. For $W = I + A$ it is $1/(1+\bar{d})$, where $\bar{d} = \sum_i \frac{t_i}{T} d_i$ is the population-weighted average degree.

Appendix D. Varying t and W

D.1. **Variable unit population, perturbed grid.** We turn to Iowa, where the largest county (Polk) is about 130 times more populous than the smallest (Adams). Iowa's county dual graph is notable because it is an approximate grid, with many squares and many triangles. The degree distribution is relatively narrow, with most interior nodes of degree 4, 5, or 6.

In each trial in Figure 10, county-level probabilities π_i are drawn from one of four families (independent across counties; high on a random independent set of counties; decaying with graph distance from one random center; or from four spread-out centers), calibrated so that the expected statewide X share is $\rho = 0.3$. Each resident is then assigned independently with probability π_i.

The equal-population control (keep experimental ρ_i but set t_i constant) shows that the weighting adjustment is modest when population heterogeneity is removed and what remains is the small variation in graph degree. With actual county populations, however, the differences become substantially larger across all three components. The effect is also distribution-dependent rather than a uniform shift: different spatial patterns can push the weighted quantities in different directions.

The magnitude of these effects can be substantial. With actual county populations and local environment $W = I + A$, the ordinary population share is essentially fixed at $\rho = 0.30$, while the pair-weighted share ρ_W ranges from about 0.16 to 0.52. This is because ρ_W is based on neighboring pairs of individuals, and these are far more likely to occur in large units near other large units.

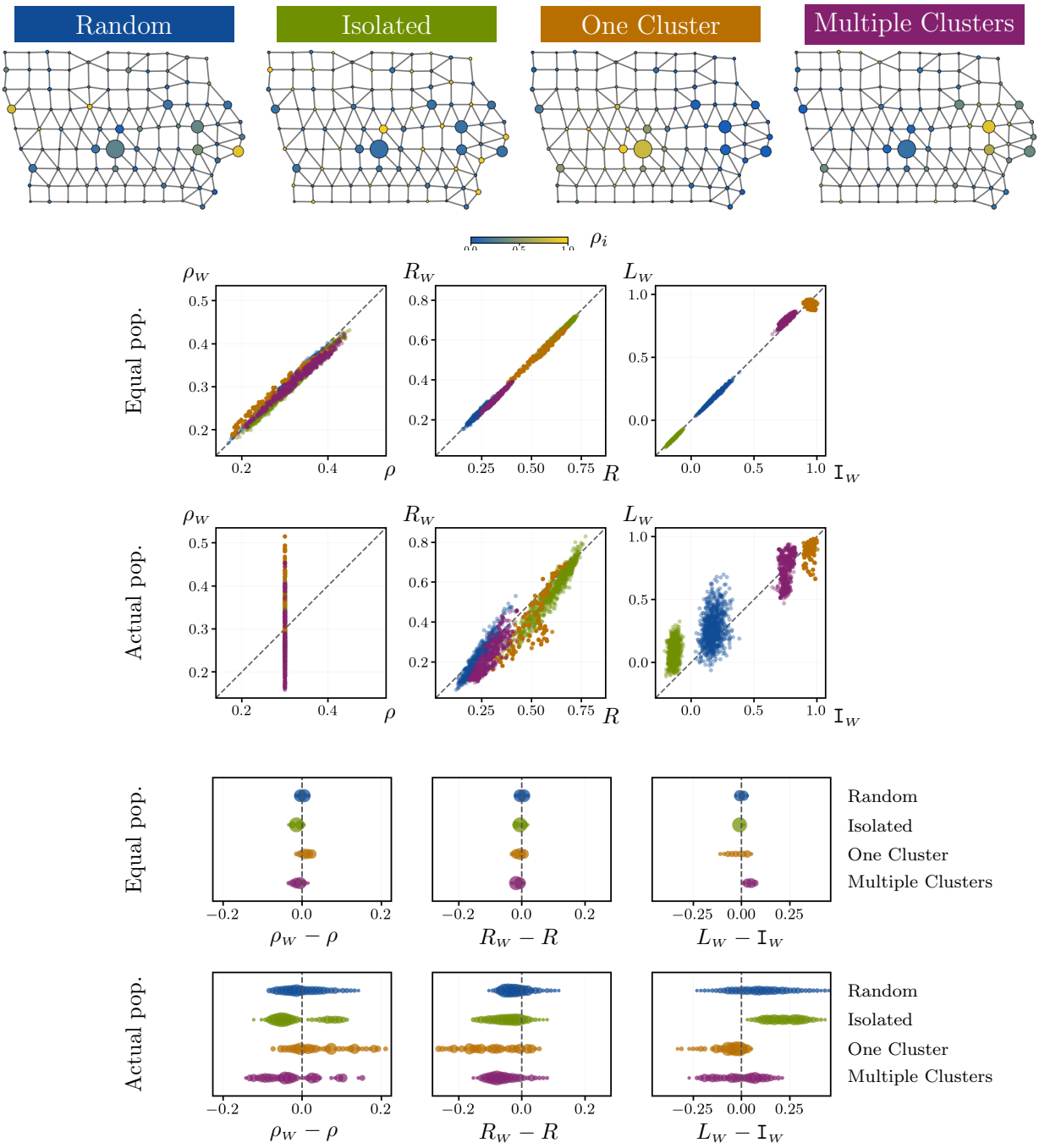


FIGURE 10. Four families of synthetic X/Y distributions are generated on the Iowa county dual graph: random, isolated, one-cluster, and multiple-cluster. The top row displays the dual graph with node area proportional to county population; coloring indicates county share ρ_i in a sample distribution targeted at $\rho = .3$ overall. The scores $\rho, R, \mathtt{I}$ and their weighted versions are compared under an equal-population treatment and with real population variation over 800 trials per family.

Thus a group concentrated in those areas can be strongly overrepresented among pair endpoints. In the same way, intensity and layout readings can be substantially different.

For random configurations, for example, $\mathtt{I}_W$ is generally near 0.2, whereas L_W ranges from slightly negative to about 0.7—a much larger span that includes values with materially different interpretations, which seems substantively correct for a randomly distributed 30% minority group.

D.2. **Spatial weighting and the local environment.** Capy can be turned into a within-unit score in the vein of evenness by setting $W = I$ instead of the standard choice $W = I + A$. And indeed we can interpolate between the two by varying a parameter $\lambda \geq 0$ in $W = I + \lambda A$. Figure 11 shows that the qualitative picture changes surprisingly little to compare $W = I$ with $W = I + A$. One conclusion is that the scale that captures maximum alternation in patterns is not the tract scale: if the city were a tract checkerboard, with high and low composition alternating by tract, we would see a huge jump in its layout score when transitioning from $W = I + A$, where L_W is negative, to $W = I$, where $L_W = 1$.

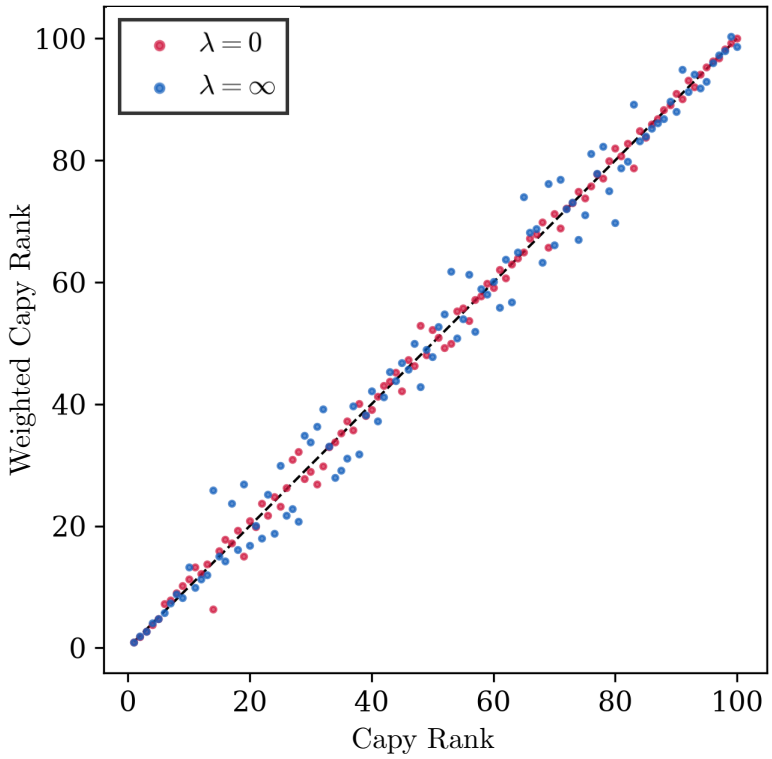


FIGURE 11. Comparison of Capy's ($\lambda = 1$) rankings of the 100 most populous U.S. metro areas to those of its weighted variants ($\lambda = 0$ and $\lambda = \infty$)

Furthermore, sending $\lambda \to \infty$, so that $W = A$ in the limit, also makes little change in the rankings. This gives us a second way of seeing that the patterns in American metros are fairly smooth at the tract level: at $W = I$, all neighbor pairs are within-tract. At $W = A$, no neighbor pairs are within tract; instead, your neighbors are only the people in the neighboring tracts. The fact that the endpoint correlation is roughly the same in either case is a statement about tract-level smoothness.

Appendix E. Data curation and construction of dual graphs

We assemble decennial Census population and geographic data at the county, tract, block-group, and block levels for 1980–2020, subject to the availability of each geography in each decade. The source data are standardized to a common set of population fields, with the principal analyses using total population, White (non-Hispanic White alone), Black (non-Hispanic Black alone), and people of color (POC), defined as total population minus White; Asian population is also used in one supplementary analysis. We use the single-race White and Black categories throughout because broader multiracial definitions are not consistently available across the full 1980–2020 time series. Historical population records are joined to contemporaneous Census boundary files by geographic identifier, with unmatched records and other exceptional cases retained for audit rather than silently treated as zero. The replication repository documents the underlying Census products, validation checks, and special handling required for early decades.

For longitudinal comparisons, we hold each metropolitan study area (MSA) fixed to its 2020 geographic footprint, so that changes in the measured quantities are not driven by changes in metropolitan extent. Within that fixed outer boundary, however, we use the Census geography native to each decade: the dual graph is rebuilt from the units and boundaries then in use, allowing tract, block group, and block splits, mergers, and boundary revisions to alter the graph over time. This avoids imposing a common historical geography through areal interpolation or crosswalk assumptions, at the cost of allowing some sensitivity to changes in Census geography. Because 1980 tracts did not cover the whole country, we impose the rule that a metro is included in the longitudinal panel only if at least 90% of its 1980 population falls in mapped 1980 tracts; this leaves $N = 280$ metros with 2020 population over 100,000. Graphs use rook adjacency (Figure 12). Units with zero combined White and Black population are removed before metric computation, while their population counts are retained separately for accounting; if this removal disconnects the graph, the surviving components are reconnected by a minimum-distance spanning tree using shortest polygon-to-polygon separation, with the added edges explicitly marked as artificial. The same artificial-edge procedure is used to join islands to the main graph.

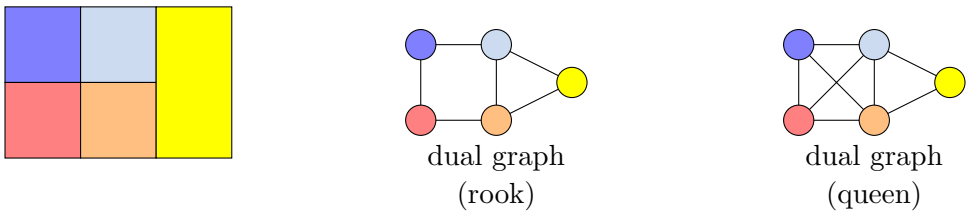


Figure 12. On the left is a partition of a region into five units. The middle and righthand figures represent dual graphs of this partition, where the middle figure has used rook adjacency and the righthand figure uses queen adjacency. With the self-pairing convention in most of the paper, the nodes would also get self-loops.

For figures that isolate a single city within each metropolitan area, we select the 2020 Census Place with the largest population inside that metropolitan area, based on 2020 census block populations. The resulting city study area is then taken to be the full 2020 boundary of that selected city.

Appendix F. More illustrations of trends

Figure 13 is designed to give a different view on the rising and falling scores through a scatterplot for all five timestamps, with a fit line for each year.

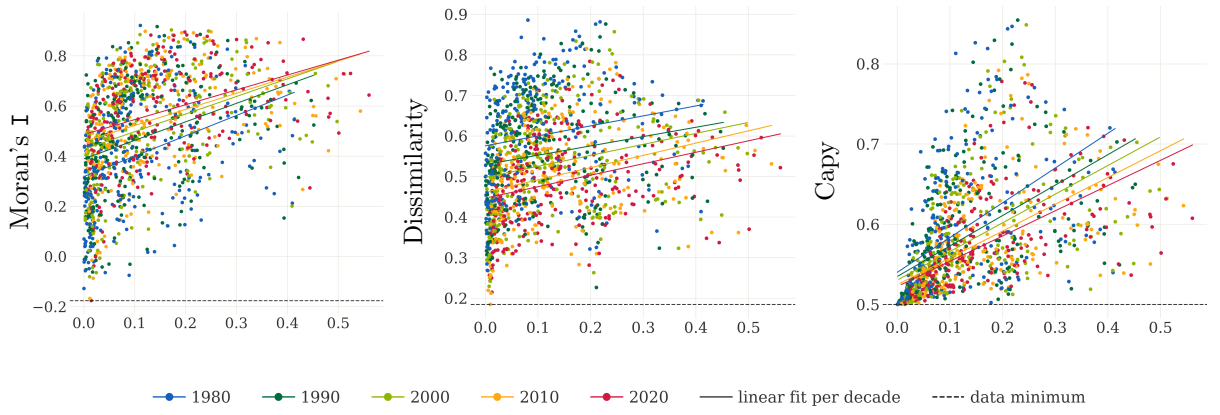


Figure 13. Another view of the trends: the metrics are plotted against ρ, the share of Black population, across Metro areas. We include metros with population over 100,000 and with data availability back to 1980 ($N = 280$).

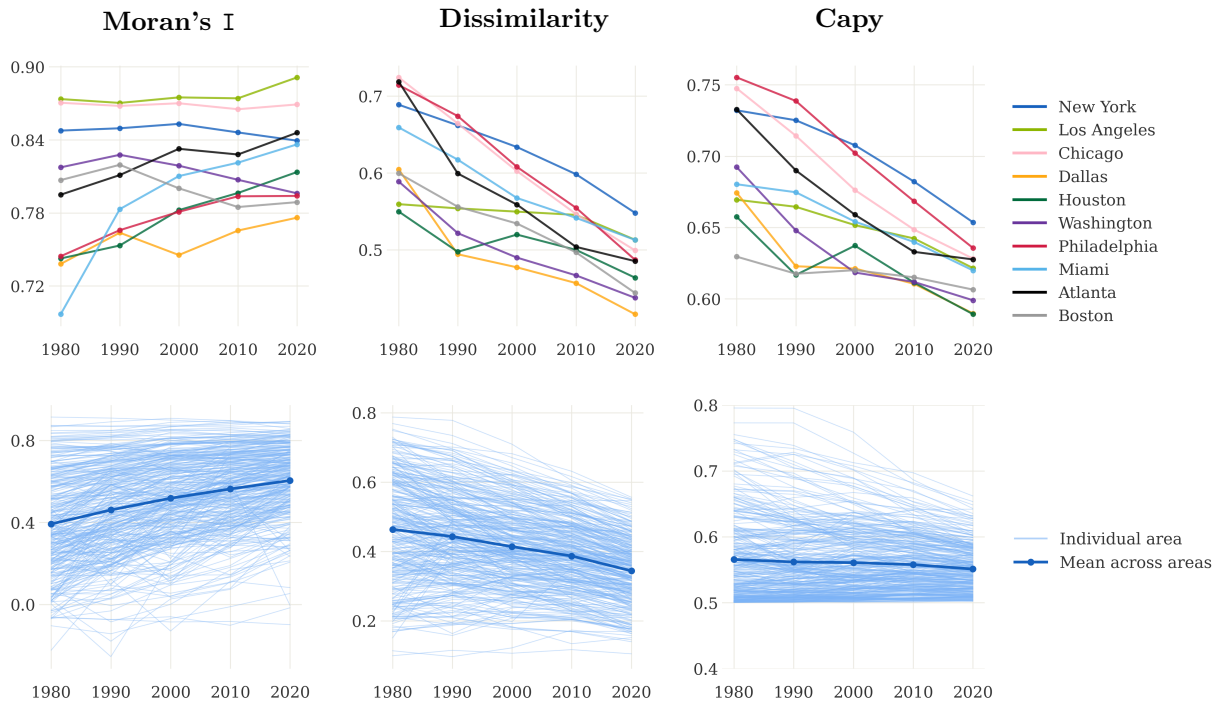


Figure 14. Switching from Black/White to **POC/White segregation** on the level of metro areas (MSAs) over time. Top trace plots show values in 10 most populous metro areas. Bottom plots show all metro areas with over 100,000 population and appearing in all decades ($N = 280$).

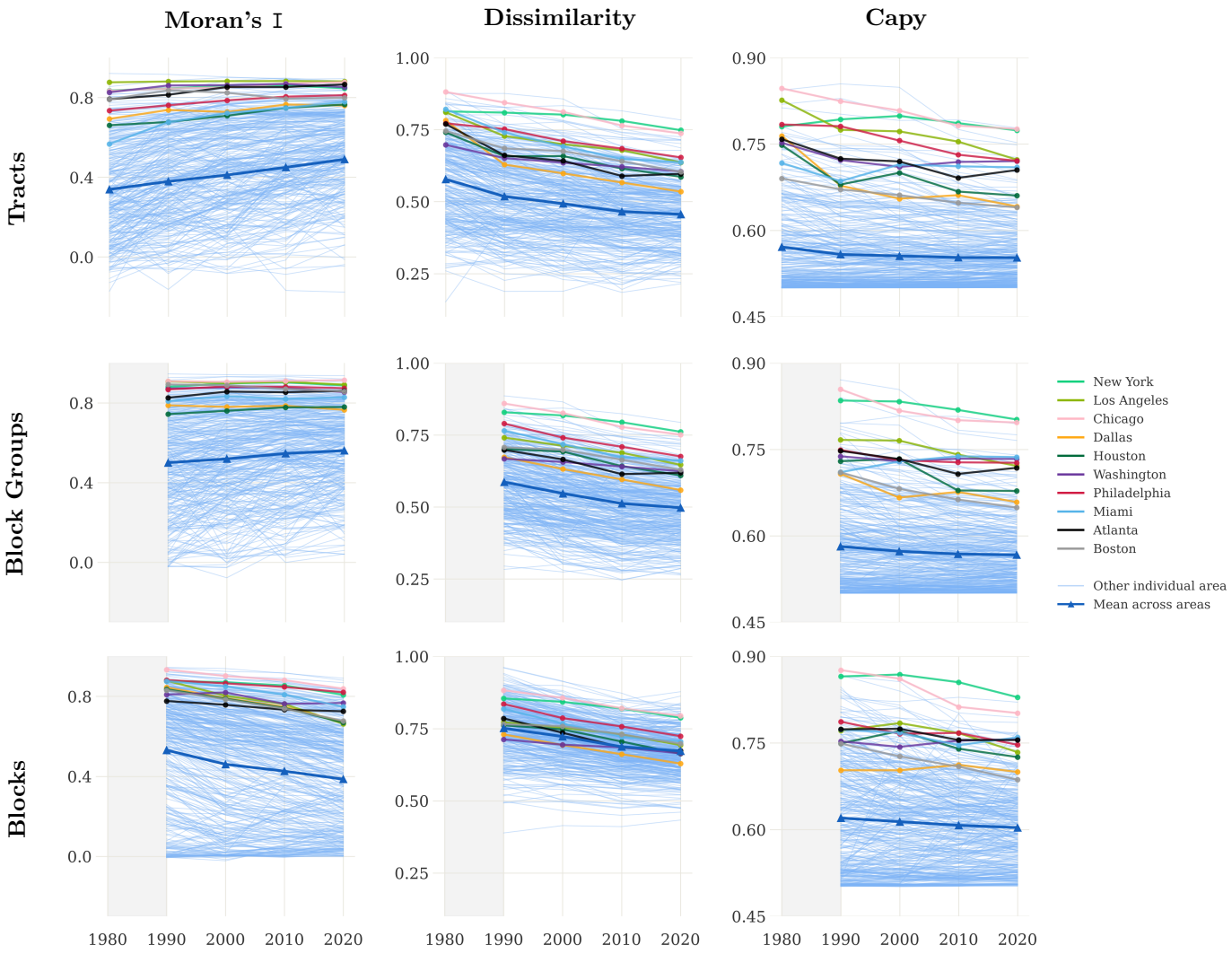


FIGURE 15. **The modifiable areal unit problem.** Black/White segregation across metro areas (MSAs) over time is shown at the level of tracts, block groups, and blocks. (Blocks and block groups are far from full coverage in 1980.) Moran's I is especially susceptible to the MAUP: the trend becomes negative on the block level. Dissimilarity also shows a collapse at the block level where the largest metros no longer sit far above the mean. Capy is the most stable.

Figures 14 and 15 show variations on our themes: switching from Black/White to POC/White shows the balancing act of Capy even more clearly; Capy is also by far best insulated from MAUP effects.

Appendix G. Observed examples in urban geography

G.1. **Hispanic and Asian population examples.** Little Village and Chinatown are two neighborhoods of Chicago that show relevant diffusive trends for other ethnic and racial groups, depicted in Figure 16.

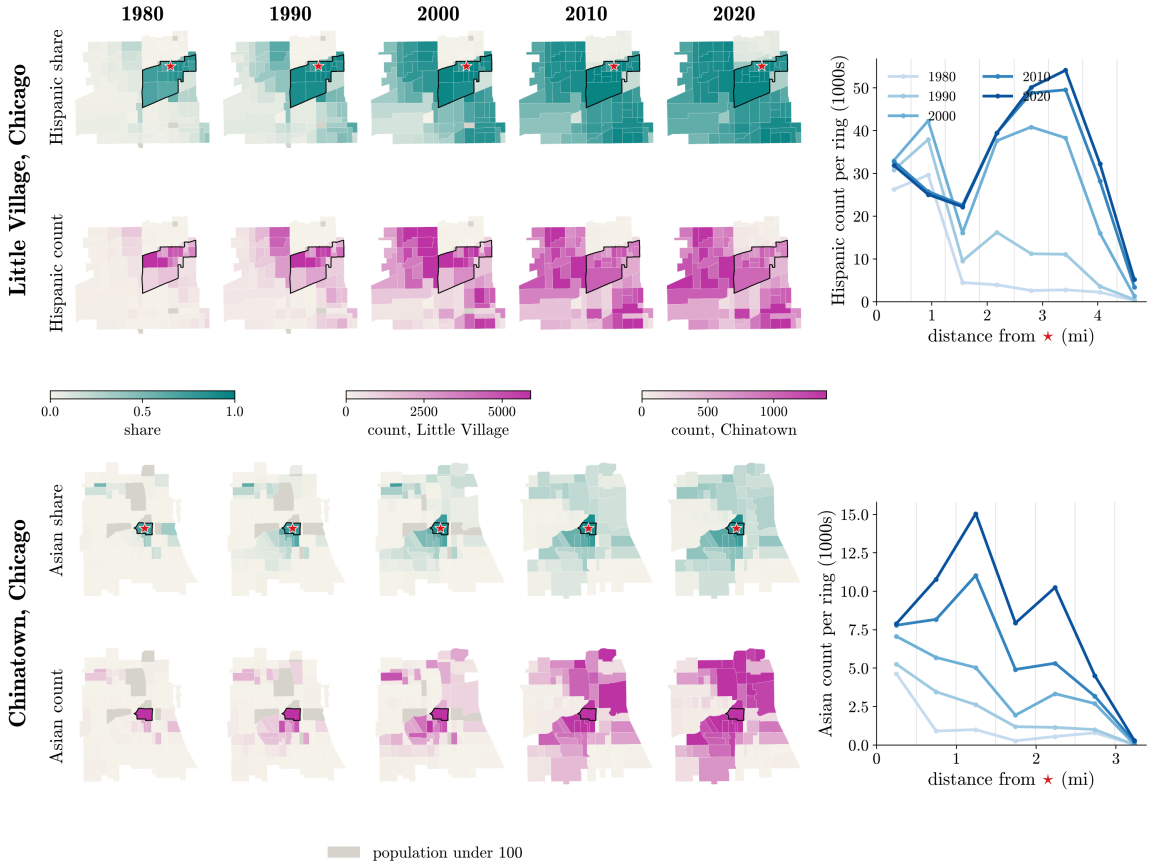


Figure 16. Diffusive patterns in two other Chicago neighborhoods: Hispanic population around Little Village, and Asian population around Chinatown. On the right, the lightest trace (1980) shows concentration near the central tract (medoid) and a radial dropoff; as time progresses, the outer areas fill in.

Sharp growth in both examples means that they compare to the growth variants 1A and 2A in Figure 4. The inner core becomes less pronounced over time in both cases.

G.2. **Tabular data.** Table 3 shows the principal measures across the top 100 MSAs by 2020 total population (TOTPOP). Metro names have been shortened by suppressing the full list of states, so for instance "Philadelphia, PA-NJ-DE-MD" lists as "Philadelphia, PA+." Recall that the universe for the Black/White comparison is Black+White population, so the Black share ρ is BPOP/BPOP+WPOP. However, the 2020 TOTPOP is reported fully, not for the restricted universe. Top ten metros by TOTPOP are shown in bold. R_W is intensity and L_W is layout, so $2C - 1 = R_W L_W$. Scores are omitted where less than 90% of the 1980 TOTPOP is covered by tracts.

TABLE 3. Black/White segregation for top 100 MSAs, ordered by C in 2020. Population in millions.

2020 Ranks			**Metro Area**	**1980 Scores**			**2000 Scores**			**2020 Scores**			**2020 Stats**	
R_W	L_W	C		R_W	L_W	C	R_W	L_W	C	R_W	L_W	C	Pop.	ρ
1	9	1	Milwaukee, WI	0.679	0.922	0.813	0.650	0.895	0.791	0.627	0.887	0.778	1.57	0.199
3	**4**	**2**	**Chicago, IL+**	**0.789**	**0.878**	**0.846**	**0.690**	**0.893**	**0.808**	**0.610**	**0.905**	**0.776**	**9.62**	**0.243**
2	**7**	**3**	**New York, NY+**	**0.656**	**0.855**	**0.780**	**0.676**	**0.885**	**0.799**	**0.611**	**0.896**	**0.774**	**20.14**	**0.256**
4	8	4	Detroit, MI	0.759	0.906	0.844	0.761	0.916	0.849	0.567	0.890	0.752	4.39	0.254
5	6	5	Cleveland, OH	0.733	0.871	0.819	0.663	0.914	0.803	0.553	0.899	0.748	2.09	0.222
8	3	6	St. Louis, MO+	0.655	0.856	0.780	0.547	0.876	0.739	0.501	0.906	0.727	2.82	0.202
10	**1**	**7**	**Los Angeles, CA**	**0.702**	**0.929**	**0.826**	**0.577**	**0.942**	**0.772**	**0.472**	**0.945**	**0.723**	**13.20**	**0.177**
7	**10**	**8**	**Philadelphia, PA+**	**0.661**	**0.859**	**0.784**	**0.585**	**0.875**	**0.756**	**0.502**	**0.880**	**0.721**	**6.25**	**0.251**
9	**2**	**9**	**Washington, DC+**	**0.552**	**0.891**	**0.746**	**0.466**	**0.902**	**0.710**	**0.485**	**0.908**	**0.720**	**6.39**	**0.362**
6	**27**	**10**	**Miami, FL**	**0.667**	**0.651**	**0.717**	**0.547**	**0.784**	**0.714**	**0.505**	**0.831**	**0.710**	**6.14**	**0.392**
13	**5**	**11**	**Atlanta, GA**	**0.563**	**0.829**	**0.733**	**0.489**	**0.900**	**0.720**	**0.454**	**0.901**	**0.705**	**6.09**	**0.431**
11	14	12	Buffalo, NY	0.626	0.769	0.741	0.575	0.841	0.742	0.467	0.859	0.701	1.17	0.146
14	25	13	Baltimore, MD	0.645	0.838	0.770	0.526	0.853	0.724	0.450	0.836	0.688	2.84	0.348
15	26	14	Birmingham, AL	0.597	0.744	0.722	0.552	0.852	0.735	0.446	0.835	0.686	1.12	0.330
16	33	15	Memphis, TN+	0.537	0.760	0.704	0.504	0.801	0.702	0.436	0.804	0.675	1.34	0.524
19	**22**	**16**	**Houston, TX**	**0.609**	**0.815**	**0.748**	**0.479**	**0.836**	**0.700**	**0.383**	**0.839**	**0.660**	**7.12**	**0.335**
17	28	17	Dayton, OH	0.583	0.823	0.740	0.501	0.835	0.709	0.386	0.829	0.660	0.81	0.176
12	73	18	New Orleans, LA	0.544	0.651	0.677	0.542	0.723	0.696	0.460	0.689	0.658	1.27	0.405
22	16	19	Hartford, CT	0.464	0.888	0.706	0.364	0.856	0.656	0.367	0.843	0.655	1.21	0.145
23	15	20	Indianapolis, IN	0.582	0.837	0.744	0.444	0.820	0.682	0.361	0.848	0.653	2.11	0.179
27	12	21	Rochester, NY	0.382	0.811	0.655	0.372	0.850	0.658	0.352	0.860	0.651	1.09	0.132
25	23	22	Columbus, OH	0.489	0.767	0.688	0.353	0.777	0.637	0.358	0.839	0.650	2.14	0.183
18	46	23	Baton Rouge, LA	—	—	—	0.420	0.753	0.658	0.385	0.770	0.648	0.87	0.392
20	45	24	Bridgeport, CT	0.359	0.596	0.607	0.371	0.682	0.627	0.382	0.775	0.648	0.96	0.153
21	42	25	Cincinnati, OH+	0.538	0.583	0.657	0.458	0.687	0.657	0.376	0.781	0.647	2.26	0.137

TABLE 3. Black/White segregation for top 100 MSAs, ordered by C in 2020. Population in millions.

2020 Ranks			**Metro Area**	**1980 Scores**			**2000 Scores**			**2020 Scores**			**2020 Stats**	
R_W	L_W	C		R_W	L_W	C	R_W	L_W	C	R_W	L_W	C	Pop.	ρ
30	18	26	Jacksonville, FL	0.584	0.793	0.732	0.347	0.802	0.639	0.339	0.843	0.643	1.61	0.259
32	17	27	Toledo, OH	0.533	0.779	0.708	0.443	0.775	0.672	0.337	0.843	0.642	0.65	0.156
31	**24**	**28**	**Dallas, TX**	**0.642**	**0.819**	**0.763**	**0.377**	**0.822**	**0.655**	**0.338**	**0.837**	**0.642**	**7.64**	**0.268**
33	**21**	**29**	**Boston, MA+**	**0.455**	**0.837**	**0.690**	**0.385**	**0.836**	**0.661**	**0.333**	**0.841**	**0.640**	**4.94**	**0.094**
24	56	30	New Haven, CT	0.392	0.682	0.633	0.394	0.755	0.649	0.358	0.745	0.634	0.86	0.178
37	19	31	San Francisco, CA	0.525	0.815	0.714	0.417	0.812	0.669	0.315	0.842	0.633	4.75	0.158
26	58	32	Jackson, MS	0.409	0.617	0.626	0.392	0.720	0.641	0.357	0.736	0.632	0.59	0.511
40	20	33	Louisville, KY+	0.512	0.777	0.699	0.418	0.779	0.663	0.312	0.841	0.631	1.29	0.169
28	55	34	Greensboro, NC	0.419	0.668	0.640	0.341	0.716	0.622	0.349	0.751	0.631	0.78	0.323
36	29	35	Orlando, FL	0.510	0.639	0.663	0.317	0.731	0.616	0.316	0.823	0.630	2.67	0.250
29	49	36	Little Rock, AR	0.427	0.589	0.626	0.368	0.672	0.624	0.343	0.759	0.630	0.75	0.268
35	43	37	Richmond, VA	0.452	0.745	0.668	0.369	0.772	0.642	0.323	0.780	0.626	1.31	0.331
38	47	38	Syracuse, NY	0.401	0.585	0.617	0.406	0.699	0.642	0.314	0.763	0.620	0.66	0.100
34	62	39	Winston, NC	0.474	0.675	0.660	0.354	0.685	0.621	0.327	0.732	0.620	0.68	0.206
47	11	40	Kansas City, MO+	0.604	0.880	0.766	0.437	0.884	0.693	0.276	0.863	0.619	2.19	0.147
39	59	41	Pittsburgh, PA	0.459	0.670	0.654	0.400	0.700	0.640	0.314	0.735	0.615	2.37	0.092
42	44	42	Charlotte, NC+	0.331	0.546	0.590	0.289	0.681	0.598	0.295	0.779	0.615	2.66	0.271
41	50	43	Columbia, SC	—	—	—	0.283	0.704	0.599	0.296	0.757	0.612	0.83	0.370
49	32	44	Tampa, FL	0.532	0.726	0.693	0.376	0.782	0.647	0.275	0.805	0.611	3.18	0.159
43	61	45	Cape Coral, FL	0.657	0.584	0.692	0.414	0.776	0.661	0.291	0.734	0.607	0.76	0.102
44	65	46	Harrisburg, PA	0.350	0.620	0.609	0.381	0.666	0.627	0.291	0.729	0.606	0.59	0.121
51	38	47	Omaha, NE+	0.383	0.742	0.642	0.384	0.802	0.654	0.266	0.786	0.604	0.97	0.095
46	60	48	Tulsa, OK	0.530	0.639	0.669	0.374	0.745	0.639	0.278	0.735	0.602	1.02	0.116
45	71	49	Virginia Beach, VA+	0.387	0.635	0.623	0.269	0.593	0.580	0.282	0.692	0.598	1.80	0.362
48	69	50	Albany, NY	0.229	0.577	0.566	0.271	0.662	0.590	0.276	0.705	0.597	0.90	0.097
52	64	51	Augusta, GA+	0.259	0.575	0.575	0.259	0.652	0.584	0.264	0.729	0.596	0.61	0.397
54	51	52	Nashville, TN	0.442	0.699	0.655	0.308	0.751	0.616	0.251	0.757	0.595	1.99	0.172
59	13	53	Denver, CO	—	—	—	0.291	0.862	0.626	0.219	0.860	0.594	2.96	0.080
50	75	54	Akron, OH	0.478	0.697	0.667	0.392	0.701	0.637	0.273	0.685	0.594	0.70	0.139
55	57	55	Springfield, MA	0.402	0.641	0.629	0.323	0.736	0.619	0.248	0.745	0.592	0.70	0.081
53	77	56	Durham, NC	—	—	—	0.243	0.620	0.575	0.260	0.655	0.585	0.65	0.320

TABLE 3. Black/White segregation for top 100 MSAs, ordered by C in 2020. Population in millions.

2020 Ranks			**Metro Area**	**1980 Scores**			**2000 Scores**			**2020 Scores**			**2020 Stats**	
R_W	L_W	C		R_W	L_W	C	R_W	L_W	C	R_W	L_W	C	Pop.	ρ
57	66	57	Oklahoma City, OK	0.489	0.787	0.693	0.293	0.707	0.604	0.232	0.728	0.585	1.43	0.145
56	67	58	Grand Rapids, MI	0.432	0.724	0.656	0.320	0.750	0.620	0.232	0.728	0.585	1.09	0.081
60	39	59	Raleigh, NC	0.260	0.682	0.589	0.172	0.692	0.560	0.211	0.785	0.583	1.41	0.235
63	36	60	Sacramento, CA	0.186	0.643	0.560	0.223	0.793	0.588	0.198	0.798	0.579	2.40	0.121
58	76	61	Wichita, KS	0.499	0.769	0.692	0.333	0.617	0.603	0.227	0.674	0.577	0.65	0.097
62	52	62	Minneapolis, MN+	0.248	0.647	0.580	0.210	0.751	0.579	0.198	0.752	0.575	3.69	0.112
67	31	63	San Antonio, TX	0.389	0.794	0.654	0.265	0.791	0.605	0.184	0.805	0.574	2.56	0.165
61	72	64	Stockton, CA	0.286	0.709	0.602	0.193	0.612	0.559	0.208	0.690	0.572	0.78	0.209
70	30	65	Las Vegas, NV	0.437	0.670	0.646	0.210	0.746	0.578	0.173	0.813	0.570	2.27	0.235
71	37	66	Riverside, CA	0.178	0.463	0.541	0.166	0.712	0.559	0.171	0.789	0.568	4.60	0.191
73	48	67	Fresno, CA	0.415	0.731	0.652	0.216	0.748	0.581	0.170	0.761	0.564	1.01	0.140
68	74	68	Knoxville, TN	0.356	0.605	0.608	0.225	0.672	0.576	0.184	0.686	0.563	0.88	0.063
72	63	69	Bakersfield, CA	0.330	0.655	0.608	0.192	0.579	0.556	0.171	0.731	0.563	0.91	0.143
64	86	70	North Port, FL	0.526	0.423	0.611	0.339	0.461	0.578	0.196	0.592	0.558	0.83	0.071
66	83	71	Poughkeepsie, NY	0.229	0.421	0.548	0.210	0.519	0.554	0.188	0.617	0.558	0.70	0.142
65	84	72	Deltona, FL	0.415	0.478	0.599	0.236	0.532	0.563	0.188	0.614	0.558	0.67	0.124
77	41	73	San Diego, CA	0.348	0.812	0.641	0.213	0.817	0.587	0.147	0.781	0.557	3.30	0.093
78	40	74	Austin, TX	0.431	0.763	0.665	0.218	0.757	0.583	0.143	0.784	0.556	2.28	0.118
79	53	75	Seattle, WA	0.221	0.693	0.577	0.142	0.696	0.549	0.135	0.751	0.551	4.02	0.093
69	93	76	Charleston, SC	0.259	0.604	0.578	0.199	0.472	0.547	0.176	0.539	0.547	0.80	0.266
81	68	77	Worcester, MA+	0.055	0.399	0.511	0.063	0.532	0.517	0.130	0.722	0.547	0.98	0.062
74	90	78	Greenville, SC	0.242	0.421	0.551	0.185	0.517	0.548	0.158	0.561	0.544	0.93	0.179
83	35	79	Phoenix, AZ	0.224	0.680	0.576	0.091	0.774	0.535	0.111	0.798	0.544	4.85	0.093
80	88	80	Palm Bay, FL	0.346	0.363	0.563	0.162	0.448	0.536	0.131	0.586	0.538	0.61	0.116
82	79	81	Providence, RI+	0.211	0.593	0.562	0.135	0.625	0.542	0.116	0.650	0.538	1.68	0.061
75	99	82	San Juan, PR	—	—	—	0.076	0.521	0.520	0.153	0.464	0.536	2.08	0.168
84	81	83	Des Moines, IA	0.272	0.564	0.577	0.168	0.653	0.555	0.109	0.633	0.534	0.71	0.066
76	100	84	Lakeland, FL	0.365	0.390	0.571	0.267	0.382	0.551	0.152	0.441	0.533	0.73	0.205
87	85	85	Allentown, PA+	0.059	0.422	0.513	0.067	0.537	0.518	0.097	0.607	0.529	0.86	0.073
85	92	86	Madison, WI	—	—	—	0.102	0.481	0.525	0.104	0.545	0.528	0.68	0.055
86	96	87	McAllen, TX	0.008	0.074	0.500	0.145	0.464	0.534	0.099	0.494	0.525	0.87	0.059

TABLE 3. Black/White segregation for top 100 MSAs, ordered by C in 2020. Population in millions.

2020 Ranks			**Metro Area**	**1980 Scores**			**2000 Scores**			**2020 Scores**			**2020 Stats**	
R_W	L_W	C		R_W	L_W	C	R_W	L_W	C	R_W	L_W	C	Pop.	ρ
88	70	88	Urban Honolulu, HI	0.102	0.539	0.528	0.099	0.612	0.530	0.068	0.703	0.524	1.02	0.099
91	54	89	Portland, OR+	0.292	0.814	0.619	0.139	0.832	0.558	0.059	0.751	0.522	2.51	0.041
92	34	90	Colorado Springs, CO	0.075	0.602	0.523	0.086	0.744	0.532	0.056	0.801	0.522	0.76	0.075
90	78	91	El Paso, TX	0.121	0.251	0.515	0.106	0.624	0.533	0.060	0.654	0.520	0.87	0.197
89	87	92	San Jose, CA	0.078	0.741	0.529	0.056	0.655	0.518	0.062	0.589	0.518	2.00	0.069
93	82	93	Tucson, AZ	0.111	0.357	0.520	0.059	0.498	0.515	0.055	0.629	0.517	1.04	0.063
94	80	94	Salt Lake City, UT	0.014	0.685	0.505	0.016	0.506	0.504	0.041	0.643	0.513	1.26	0.025
95	89	95	Albuquerque, NM	—	—	—	0.042	0.511	0.511	0.038	0.577	0.511	0.92	0.058
96	94	96	Oxnard, CA	0.091	0.556	0.525	0.053	0.642	0.517	0.030	0.532	0.508	0.84	0.037
97	95	97	Boise City, ID	0.005	0.296	0.501	0.002	0.536	0.501	0.020	0.515	0.505	0.76	0.016
98	91	98	Spokane, WA	0.035	0.449	0.508	0.021	0.494	0.505	0.015	0.561	0.504	0.59	0.022
99	97	99	Ogden, UT	0.133	0.160	0.511	0.032	0.309	0.505	0.010	0.478	0.502	0.69	0.013
100	98	100	Provo, UT	0.002	0.244	0.500	0.002	0.456	0.500	0.004	0.472	0.501	0.67	0.008

We have chosen to focus on the largest metros by total population, but sometimes the study universe is much smaller. If restricting to B+W, the top ten list is only altered by Detroit replacing Miami, but the metros would be substantially reordered.

TABLE 4. 2020 total population and Black+White analysis population (millions).

Metro area	TOTPOP	BPOP+WPOP
New York	20.14	11.95
Los Angeles	13.20	4.61
Chicago	9.62	6.40
Dallas	7.64	4.49
Houston	7.12	3.64
Washington	6.39	4.27
Philadelphia	6.25	4.96
Miami	6.14	2.98
Atlanta	6.09	4.71
Boston	4.94	3.66
Detroit	4.39	3.76